CHAPTER

4

Differential Amplifiers

The differential amplifier is among the most important circuit inventions, dating back to the vacuum tube era. Offering many useful properties, differential operation has become the de facto choice in today's high-performance analog and mixed-signal circuits.

This chapter deals with the analysis and design of CMOS differential amplifiers. Following a review of single-ended and differential operation, we describe the basic differential pair and analyze both the large-signal and the small-signal behavior. Next, we introduce the concept of common-mode rejection and formulate it for differential amplifiers. We then study differential pairs with diode-connected and current-source loads as well as differential cascode stages. Finally, we describe the Gilbert cell.

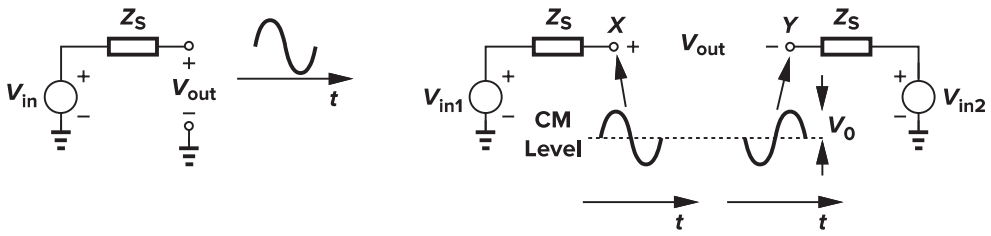
4.1 ■ Single-Ended and Differential Operation

A "single-ended" signal is defined as one that is measured with respect to a fixed potential, usually the ground [Fig. 4.1(a)]. A differential signal is defined as one that is measured between two nodes that have *equal* and *opposite* signal excursions around a fixed potential [Fig. 4.1(b)]. In the strict sense, the two nodes must also exhibit equal impedances to that potential. The "center" potential in differential signaling is called the "common-mode" (CM) level. It is helpful to view the CM level as the bias value of the voltages, i.e., the value in the absence of signals.

The specification of signal swings in a differential system can be confusing. Suppose each single-ended output in Fig. 4.1(b) has a peak amplitude of V_0. Then, the single-ended peak-to-peak swing is $2V_0$ and the differential peak-to-peak swing is $4V_0$. For example, if the voltage at X (with respect to ground) is $V_0 \cos \omega t + V_{CM}$ and that at Y is $-V_0 \cos \omega t + V_{CM}$, then the peak-to-peak swing of $V_X - V_Y$ ($=2V_0 \cos \omega t$) is $4V_0$. It is therefore not surprising that a circuit with a supply voltage of 1 V can deliver a peak-to-peak differential swing of 1.6 V.

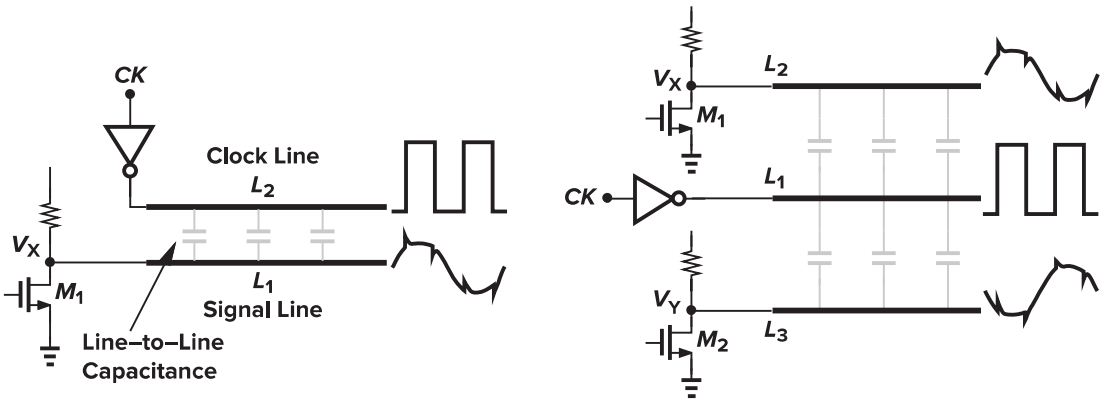
An important advantage of differential operation over single-ended signaling is higher immunity to "environmental" noise. Consider the example depicted in Fig. 4.2, where two adjacent lines in a circuit carry a small, sensitive signal and a large clock waveform. Due to capacitive coupling between the lines, transitions on line L_2 corrupt the signal on line L_1. Now suppose, as shown in Fig. 4.2(b), the sensitive signal is distributed as two equal and opposite phases. If the clock line is placed midway between the two, the transitions disturb the differential phases by equal amounts, leaving the *difference* intact. Since the common-mode level of the two phases is disturbed, but the differential output is not corrupted, we say that this arrangement "rejects" common-mode noise.[1]

[1] It is also possible to place a "shield" line between the sensitive line and the clock line (Chapter 19).



(a) (b)

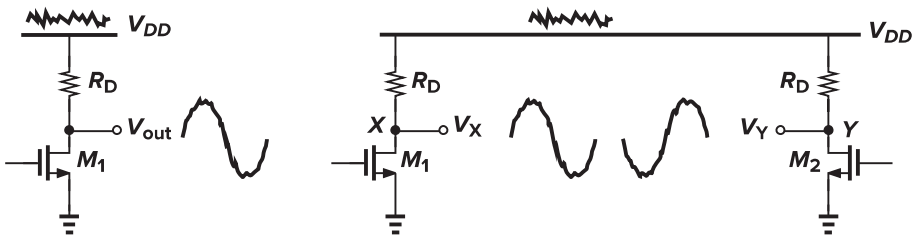
Figure 4.1 (a) Single-ended and (b) differential signals.



(a) (b)

Figure 4.2 (a) Corruption of a signal due to coupling; (b) reduction of coupling by differential operation.

Another example of common-mode rejection occurs with noisy supply voltages. In the CS stage of Fig. 4.3(a), if V_{DD} varies by ΔV, then V_{out} changes by approximately the same amount, i.e., the output is quite susceptible to noise on V_{DD}. Now consider the circuit in Fig. 4.3(b). Here, if the circuit is symmetric, noise on V_{DD} affects V_X and V_Y, but not $V_X - V_Y = V_{out}$. Thus, the circuit of Fig. 4.3(b) is much more robust in dealing with supply noise.



(a) (b)

Figure 4.3 Effect of supply noise on (a) a single-ended circuit and (b) a differential circuit.

Thus far, we have seen the importance of employing differential paths for sensitive signals ("victims"). It is also beneficial to employ differential distribution for *noisy lines* ("aggressors"). For example, suppose the clock signal of Fig. 4.2 is distributed in differential form on two lines (Fig. 4.4). Then, with perfect symmetry, the components coupled from CK and $\overline{CK}$ to the signal line cancel each other.

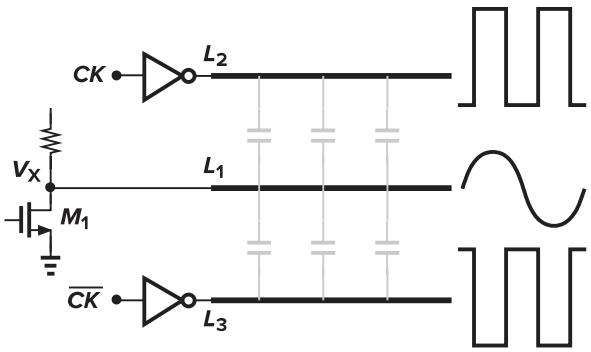


Figure 4.4 Reduction of coupled noise by differential operation.

▶ **Example 4.1**

If differential victims or differential aggressors improve the overall noise immunity, can we choose differential phases for *both* victims and aggressors?

Solution

Yes, we can. Let us consider the arrangement shown in Fig. 4.5(a), where the differential victims are surrounded by the differential aggressors. Unfortunately, in this case, $V_{out}^{+} - V_{out}^{-}$ is corrupted because V_{out}^{+} and V_{out}^{-} experience *opposite* jumps.

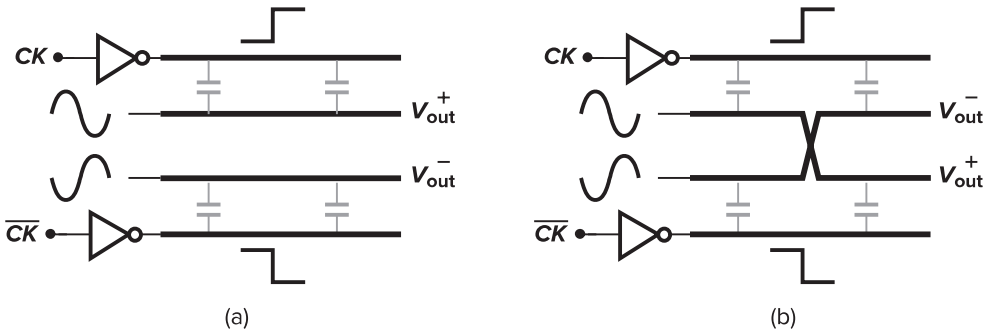


Figure 4.5

Now, suppose we modify the routing as depicted in Fig. 4.5(b), where V_{out}^{+} (V_{out}^{-}) is adjacent to CK ($\overline{CK}$) for half of the distance and to $\overline{CK}$ (CK) for the other half. In this case, the couplings from CK and $\overline{CK}$ cancel each other. Interestingly, V_{out}^{+} and V_{out}^{-} are free from the coupling—and so is their difference. This geometry is an example of "twisted pairs."

◀

Another useful property of differential signaling is the increase in maximum achievable voltage swings. In the circuit of Fig. 4.3, for example, the maximum output swing at X or Y is equal to $V_{DD} - (V_{GS} - V_{TH})$, whereas for $V_X - V_Y$, the peak-to-peak swing is equal to $2[V_{DD} - (V_{GS} - V_{TH})]$. Other advantages of differential circuits over their single-ended counterparts include simpler biasing and higher linearity (Chapter 14).

While differential circuits may occupy about twice as much area as single-ended alternatives, in practice this is a minor drawback. The numerous advantages of differential operation by far outweigh the possible increase in the area.

4.2 ■ Basic Differential Pair

How do we amplify a differential signal? As suggested by the observations in the previous section, we may incorporate two identical single-ended signal paths to process the two phases [Fig. 4.6(a)]. Here, two differential inputs, V_{in1} and V_{in2}, having a certain CM level, $V_{in,CM}$, are applied to the gates. The outputs are also differential and swing around the output CM level, $V_{out,CM}$. Such a circuit indeed offers some of the advantages of differential signaling: high rejection of supply noise, higher output swings, etc. But what happens if V_{in1} and V_{in2} experience a large common-mode disturbance or simply do not have a well-defined common-mode dc level? As the input CM level, $V_{in,CM}$, changes, so do the bias currents of M_1 and M_2, thus varying both the transconductance of the devices and the output CM level. The variation of the transconductance, in turn, leads to a change in the small-signal gain, while the departure of the output CM level from its ideal value lowers the maximum allowable output swings. For example, as shown in Fig. 4.6(b), if the input CM level is excessively low, the minimum values of V_{in1} and V_{in2} may in fact turn off M_1 and M_2, leading to severe clipping at the output. Thus, it is important that the bias currents of the devices have minimal dependence on the input CM level.

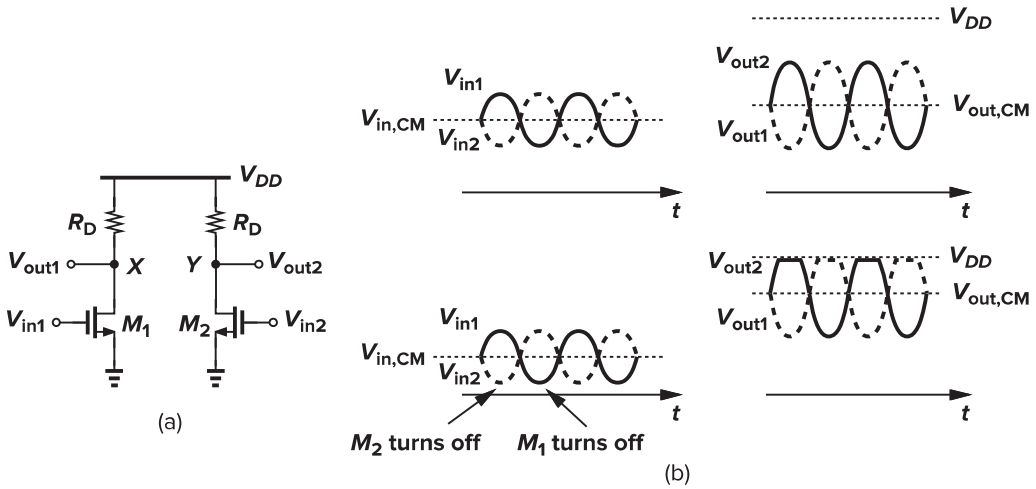


Figure 4.6 (a) Simple differential circuit; (b) illustration of sensitivity to the input common-mode level.

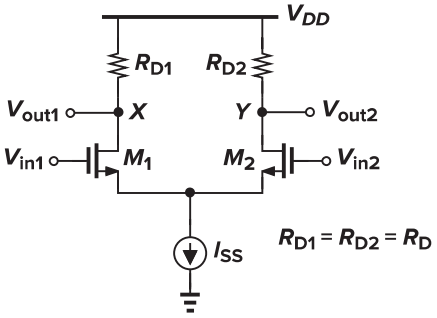


Figure 4.7 Basic differential pair.

A simple modification can resolve the above issue. Shown in Fig. 4.7, the "differential pair"[2] employs a current source I_{SS} to make $I_{D1} + I_{D2}$ independent of $V_{in,CM}$. Thus, if $V_{in1} = V_{in2}$, the bias current of

[2] Also called a "source-coupled" pair or (in the British literature) a "long-tailed" pair.

each transistor equals $I_{SS}/2$ and the output common-mode level is $V_{DD} - R_D I_{SS}/2$. It is instructive to study the large-signal behavior of the circuit for both differential and common-mode input variations. In the large-signal study, we neglect channel-length modulation and body effect.

4.2.1 Qualitative Analysis

Let us assume that in Fig. 4.7, $V_{in1} - V_{in2}$ varies from $-\infty$ to $+\infty$. If V_{in1} is much more negative than V_{in2}, M_1 is off, M_2 is on, and $I_{D2} = I_{SS}$. Thus, $V_{out1} = V_{DD}$ and $V_{out2} = V_{DD} - R_D I_{SS}$. As V_{in1} is brought closer to V_{in2}, M_1 gradually turns on, drawing a fraction of I_{SS} from R_{D1} and hence lowering V_{out1}. Since $I_{D1} + I_{D2} = I_{SS}$, the drain current of M_2 falls and V_{out2} rises. As shown in Fig. 4.8(a), for $V_{in1} = V_{in2}$, we have $V_{out1} = V_{out2} = V_{DD} - R_D I_{SS}/2$, which is the output CM level. As V_{in1} becomes more positive than V_{in2}, M_1 carries a greater current than does M_2 and V_{out1} drops below V_{out2}. For sufficiently large $V_{in1} - V_{in2}$, M_1 "hogs" all of I_{SS}, turning M_2 off. As a result, $V_{out1} = V_{DD} - R_D I_{SS}$ and $V_{out2} = V_{DD}$. Figure 4.8 also plots $V_{out1} - V_{out2}$ versus $V_{in1} - V_{in2}$. Note that the circuit contains three differential quantities: $V_{in1} - V_{in2}$, $V_{out1} - V_{out2}$, and $I_{D1} - I_{D2}$.

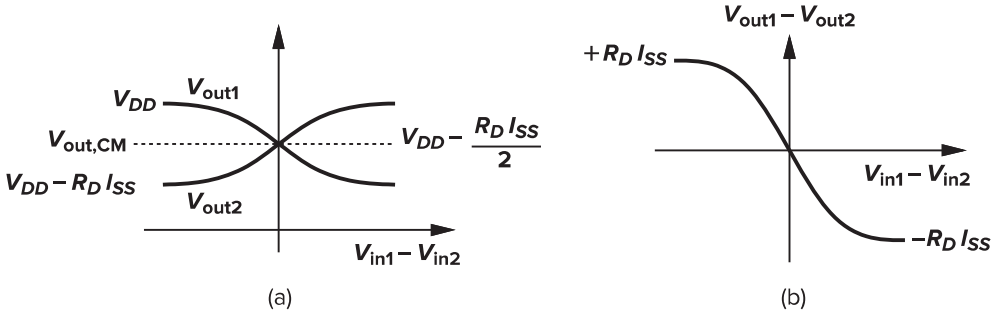


Figure 4.8 Differential input-output characteristics of a differential pair.

The foregoing analysis reveals two important attributes of the differential pair. First, the maximum and minimum levels at the output are well-defined (V_{DD} and $V_{DD} - R_D I_{SS}$, respectively) and independent of the input CM level. Second, as proved later, the small-signal gain (the slope of $V_{out1} - V_{out2}$ versus $V_{in1} - V_{in2}$) is maximum for $V_{in1} = V_{in2}$, gradually falling to zero as $|V_{in1} - V_{in2}|$ increases. In other words, the circuit becomes more nonlinear as the input voltage swing increases. For $V_{in1} = V_{in2}$, we say that the circuit is in "equilibrium."

Now let us consider the common-mode behavior of the circuit. As mentioned earlier, the role of the tail current source is to suppress the effect of input CM level variations on the operation of M_1 and M_2 and the output level. Does this mean that $V_{in,CM}$ can assume arbitrarily low or high values? To answer this question, we set $V_{in1} = V_{in2} = V_{in,CM}$ and vary $V_{in,CM}$ from 0 to V_{DD}. Figure 4.9(a) shows the circuit with I_{SS} implemented by an NFET. Note that the symmetry of the pair requires that $V_{out1} = V_{out2}$.

What happens if $V_{in,CM} = 0$? Since the gate potential of M_1 and M_2 is not more positive than their source potential, both devices are off, yielding $I_{D3} = 0$. This indicates that M_3 operates in the deep triode region because V_b is high enough to create an inversion layer in the transistor. With $I_{D1} = I_{D2} = 0$, the circuit is incapable of signal amplification, $V_{out1} = V_{out2} = V_{DD}$, and $V_P = 0$.

Now suppose $V_{in,CM}$ becomes more positive. Modeling M_3 by a resistor as in Fig. 4.9(b), we note that M_1 and M_2 turn on if $V_{in,CM} \geq V_{TH}$. Beyond this point, I_{D1} and I_{D2} continue to increase, and V_P also rises [Fig. 4.9(c)]. In a sense, M_1 and M_2 constitute a source follower, forcing V_P to track $V_{in,CM}$. For a sufficiently high $V_{in,CM}$, the drain-source voltage of M_3 exceeds $V_{GS3} - V_{TH3}$, allowing the device to operate in saturation. The total current through M_1 and M_2 then remains constant. We conclude that for proper operation, $V_{in,CM} \geq V_{GS1} + (V_{GS3} - V_{TH3})$.

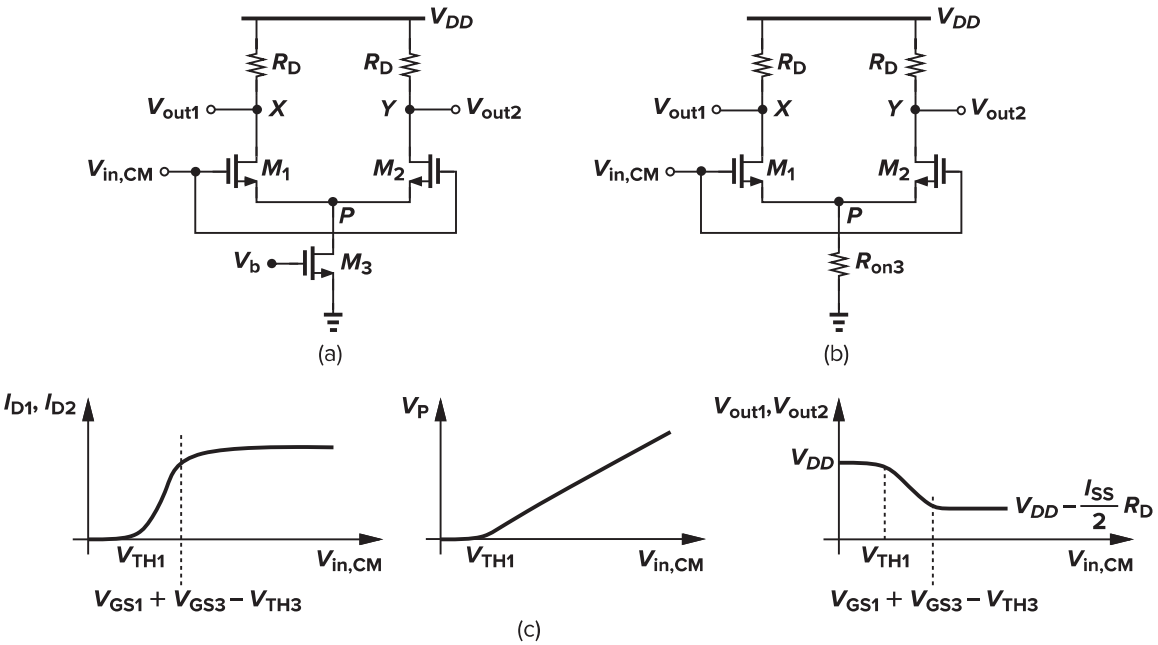


Figure 4.9 (a) Differential pair sensing an input common-mode change; (b) equivalent circuit if M_3 operates in the deep triode region; (c) common-mode input-output characteristics.

What happens if $V_{in,CM}$ rises further? Since V_{out1} and V_{out2} are relatively constant, we expect that M_1 and M_2 enter the triode region if $V_{in,CM} > V_{out1} + V_{TH} = V_{DD} - R_D I_{SS}/2 + V_{TH}$. This sets an upper limit on the input CM level. In summary, the allowable value of $V_{in,CM}$ is bounded as follows:

$$V_{GS1} + (V_{GS3} - V_{TH3}) \leq V_{in,CM} \leq \min\left[V_{DD} - R_D\frac{I_{SS}}{2} + V_{TH},\ V_{DD}\right] \tag{4.1}$$

Beyond the upper bound, the CM characteristics of Fig. 4.9(c) do not change, but the differential gain drops.[3]

▶ Example 4.2

Sketch the small-signal differential gain of a differential pair as a function of the input CM level.

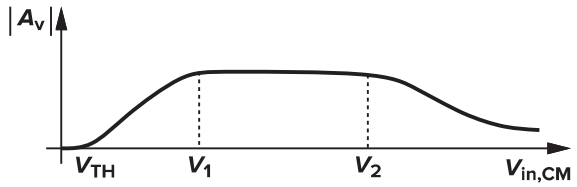


Figure 4.10

Solution

As shown in Fig. 4.10, the gain begins to increase as $V_{in,CM}$ exceeds V_{TH}. After the tail current source enters saturation ($V_{in,CM} = V_1$), the gain remains relatively constant. Finally, if $V_{in,CM}$ is so high that the input transistors enter the triode region ($V_{in,CM} = V_2$), the gain begins to fall. ◀

[3]This bound assumes small differential swings at the input and the output. This point become clear later.

With our understanding of differential and common-mode behavior of the differential pair, we can now answer another important question: How large can the output voltage swings of a differential pair be? Suppose the circuit is biased with input and output bias levels $V_{in,CM}$ and $V_{out,CM}$, respectively, and $V_{in,CM} < V_{out,CM}$. Also, assume that the voltage gain is high, that is, the input swing is much less than the output swing. As illustrated in Fig. 4.11, for M_1 and M_2 to be saturated, each output can go as high as V_{DD} but as low as approximately $V_{in,CM} - V_{TH}$. In other words, the higher the input CM level, the smaller the allowable output swings. For this reason, it is desirable to choose a relatively low $V_{in,CM}$, but, of course, no less than $V_{GS1} + (V_{GS3} - V_{TH3})$. Such a choice affords a single-ended peak-to-peak output swing of $V_{DD} - (V_{GS1} - V_{TH1}) - (V_{GS3} - V_{TH3})$ (why?). The reader is encouraged to repeat this analysis if the voltage gain is around unity.

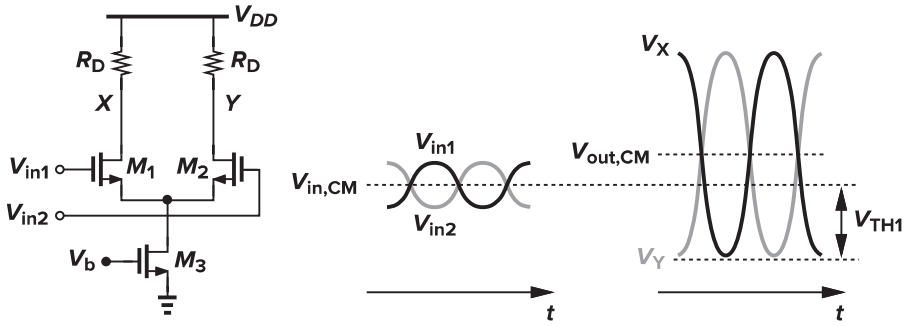


Figure 4.11 Maximum allowable output swings in a differential pair.

▶ **Example 4.3**

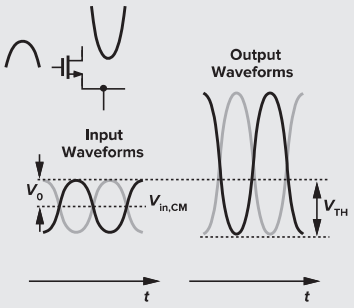
Compare the maximum output voltage swings provided by a CS stage and a differential pair.

Solution

Recall from Chapter 3 that a CS stage (with resistive load) allows an output swing of V_{DD} minus one overdrive ($V_{DD} - V_{D,sat}$). As seen above, with proper choice of the input CM level, a differential pair provides a maximum output swing of V_{DD} minus two overdrives (single-ended) or $2V_{DD}$ minus four overdrives (differential) ($2V_{DD} - 4V_{D,sat}$), which is typically quite a lot larger than $V_{DD} - V_{D,sat}$. ◀

Nanometer Design Notes

Owing to both severe channel-length modulation and limited supply voltages, the voltage gain of nanometer differential pairs hardly exceeds 5. In this case, the peak *input* swing also limits the output swing. As shown below, for a peak input amplitude of V_0, the minimum allowable output is equal to $V_{in,CM} + V_0 - V_{TH}$. This issue arises in any circuit that has a *negative* gain.



4.2.2 Quantitative Analysis

In this section, we quantify both large-signal and small-signal characteristics of MOS differential pairs. We begin with large-signal analysis to arrive at expressions for the plots shown in Fig. 4.8.

Large-Signal Behavior Consider the differential pair shown in Fig. 4.12. Our objective is to determine $V_{out1} - V_{out2}$ as a function of $V_{in1} - V_{in2}$. We have $V_{out1} = V_{DD} - R_{D1}I_{D1}$ and $V_{out2} = V_{DD} - R_{D2}I_{D2}$, that is, $V_{out1} - V_{out2} = R_{D2}I_{D2} - R_{D1}I_{D1} = R_D(I_{D2} - I_{D1})$ if $R_{D1} = R_{D2} = R_D$. Thus, we simply calculate I_{D1} and I_{D2} in terms of V_{in1} and V_{in2}, assuming the circuit is symmetric, M_1 and M_2 are saturated, and $\lambda = 0$. Since the voltage at node P is equal to $V_{in1} - V_{GS1}$ and $V_{in2} - V_{GS2}$,

$$V_{in1} - V_{in2} = V_{GS1} - V_{GS2} \tag{4.2}$$

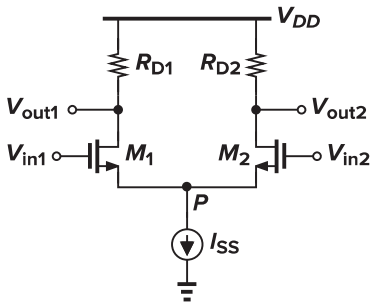


Figure 4.12 Differential pair.

For a square-law device, we have

$$(V_{GS} - V_{TH})^2 = \frac{I_D}{\frac{1}{2}\mu_n C_{ox}\frac{W}{L}} \tag{4.3}$$

and, therefore,

$$V_{GS} = \sqrt{\frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}} + V_{TH} \tag{4.4}$$

It follows from (4.2) and (4.4) that

$$V_{in1} - V_{in2} = \sqrt{\frac{2I_{D1}}{\mu_n C_{ox}\frac{W}{L}}} - \sqrt{\frac{2I_{D2}}{\mu_n C_{ox}\frac{W}{L}}} \tag{4.5}$$

We wish to calculate the differential output current, $I_{D1} - I_{D2}$. Squaring the two sides of (4.5) and recognizing that $I_{D1} + I_{D2} = I_{SS}$, we obtain

$$(V_{in1} - V_{in2})^2 = \frac{2}{\mu_n C_{ox}\frac{W}{L}}(I_{SS} - 2\sqrt{I_{D1}I_{D2}}) \tag{4.6}$$

That is,

$$\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})^2 - I_{SS} = -2\sqrt{I_{D1}I_{D2}} \tag{4.7}$$

Squaring the two sides again and noting that $4I_{D1}I_{D2} = (I_{D1}+I_{D2})^2-(I_{D1}-I_{D2})^2 = I_{SS}^2-(I_{D1}-I_{D2})^2$, we arrive at

$$(I_{D1} - I_{D2})^2 = -\frac{1}{4}\left(\mu_n C_{ox}\frac{W}{L}\right)^2 (V_{in1} - V_{in2})^4 + I_{SS}\mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})^2 \tag{4.8}$$

Thus,

$$I_{D1} - I_{D2} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})\sqrt{\frac{4I_{SS}}{\mu_n C_{ox}\frac{W}{L}} - (V_{in1} - V_{in2})^2} \tag{4.9}$$

$$= \sqrt{\mu_n C_{ox}\frac{W}{L}I_{SS}}(V_{in1} - V_{in2})\sqrt{1 - \frac{\mu_n C_{ox}(W/L)}{4I_{SS}}(V_{in1} - V_{in2})^2} \tag{4.10}$$

We can say that M_1, M_2, and the tail operate as a voltage-dependent current source producing $I_{D1} - I_{D2}$ according to the above large-signal characteristics. As expected, $I_{D1}-I_{D2}$ is an odd function of $V_{in1}-V_{in2}$,

falling to zero for $V_{in1} = V_{in2}$. As $|V_{in1} - V_{in2}|$ increases from zero, $|I_{D1} - I_{D2}|$ increases because the factor preceding the square root rises more rapidly than the argument in the square root drops.[4]

Before examining (4.9) further, it is instructive to calculate the slope of the characteristic, i.e., the equivalent G_m of M_1 and M_2. Denoting the differential quantities $I_{D1} - I_{D2}$ and $V_{in1} - V_{in2}$ by ΔI_D and ΔV_{in}, respectively, the reader can show that

$$\frac{\partial \Delta I_D}{\partial \Delta V_{in}} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\frac{\dfrac{4I_{SS}}{\mu_n C_{ox} W/L} - 2\Delta V_{in}^2}{\sqrt{\dfrac{4I_{SS}}{\mu_n C_{ox} W/L} - \Delta V_{in}^2}} \tag{4.11}$$

For $\Delta V_{in} = 0$, G_m is maximum (why?) and equal to $\sqrt{\mu_n C_{ox}(W/L) I_{SS}}$. Moreover, since $V_{out1} - V_{out2} = R_D \Delta I = -R_D G_m \Delta V_{in}$, we can write the small-signal differential voltage gain of the circuit in the equilibrium condition as

$$|A_v| = \sqrt{\mu_n C_{ox}\frac{W}{L} I_{SS}}\, R_D \tag{4.12}$$

Since each transistor carries a bias current of $I_{SS}/2$ in this condition, the factor $\sqrt{\mu_m C_{ox}(W/L) I_{SS}}$ is in fact the same as the transconductance of each device, that is, $|A_v| = g_m R_D$. Equation (4.11) also suggests that G_m falls to zero for $\Delta V_{in} = \sqrt{2I_{SS}/(\mu_n C_{ox} W/L)}$. As we will see below, this value of ΔV_{in} plays an important role in the operation of the circuit.

Let us now examine Eq. (4.9) more closely. If $(V_{in1} - V_{in2})^2 \ll 4I_{SS}/[\mu_n C_{ox}(W/L)]$, then

$$I_{D1} - I_{D2} = \sqrt{\mu_n C_{ox}\frac{W}{L} I_{SS}}\,(V_{in1} - V_{in2}) \tag{4.13}$$

which yields the same equilibrium G_m as that obtained above.

But what happens for larger values of $|V_{in1} - V_{in2}|$? It appears that the argument in the square root drops to zero for $\Delta V_{in} = \sqrt{4I_{SS}/(\mu_n C_{ox} W/L)}$ and ΔI_D crosses zero at *two* different values of ΔV_{in}, an effect not predicted by our qualitative analysis in Fig. 4.8. This conclusion, however, is incorrect. To understand why, recall that (4.9) was derived with the assumption that both M_1 and M_2 are on. In reality, as ΔV_{in} exceeds a limit, one transistor carries the entire I_{SS}, turning off the other.[5] Denoting this value by ΔV_{in1}, we have $I_{D1} = I_{SS}$ and $\Delta V_{in1} = V_{GS1} - V_{TH}$ because M_2 is nearly off. It follows that

$$\Delta V_{in1} = \sqrt{\frac{2I_{SS}}{\mu_n C_{ox}\dfrac{W}{L}}} \tag{4.14}$$

For $\Delta V_{in} > \Delta V_{in1}$, M_2 is off and (4.9) and (4.10) do not hold. As mentioned above, G_m falls to zero for $\Delta V_{in} = \Delta V_{in1}$. Figure 4.13 plots the behavior.

▶ Example 4.4

Plot the output currents of a differential pair versus ΔV_{in} as the device width and the tail current vary.

Solution

Consider the characteristic shown in Fig. 4.14(a). As W/L increases, ΔV_{in1} decreases, narrowing the input range across which both devices are on [Fig. 4.14(b)]. As I_{SS} increases, both the input range and the output current swing increase [Fig. 4.14(c)]. Intuitively, we expect the circuit to become more linear as I_{SS} increases or W/L decreases. ◀

[4] It is interesting to note that, even though I_{D1} and I_{D2} are *square* functions of their respective gate-source voltages, $I_{D1} - I_{D2}$ is an odd function of $V_{in1} - V_{in2}$. This effect is studied in Chapter 14.

[5] We neglect subthreshold conduction here.

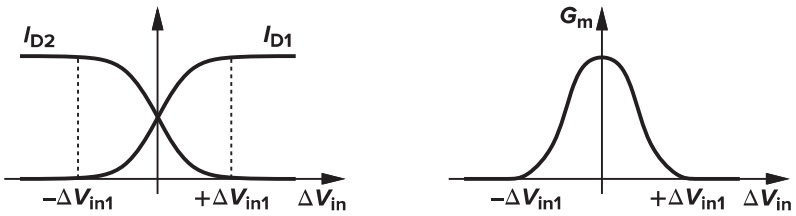


Figure 4.13 Variation of drain currents and overall transconductance of a differential pair versus input voltage.

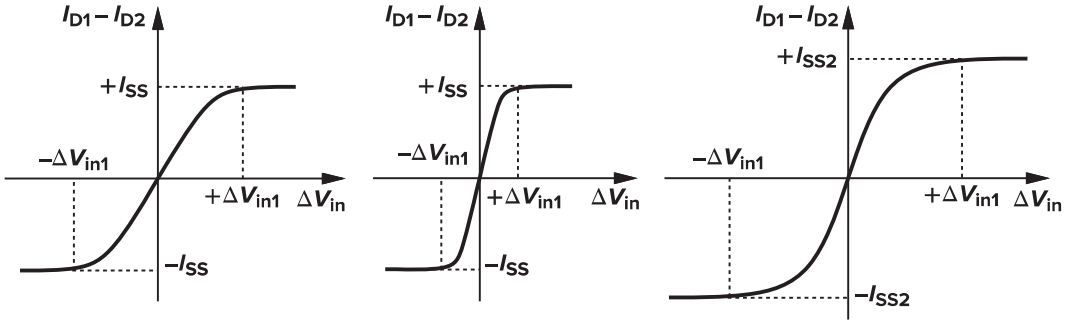
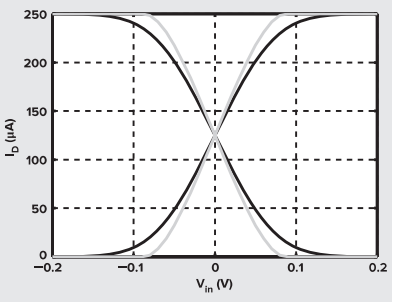


Figure 4.14

Nanometer Design Notes

Nanometer differential pairs exhibit a similar relation between the equilibrium overdrive and the differential voltage necessary to turn one side off. Plotted below are the output currents of a differential pair with $W/L = 5\ \mu$m/40 nm and $I_{SS} = 0.25$ mA using actual models (black curve) and a square-law model (gray curve). If we define cut-off as when one transistor carries 90% of the tail current, then the nanometer design also displays approximately a factor of $\sqrt{2}$ between the equilibrium and cut-off voltages.



The value of ΔV_{in1} given by (4.14) in essence represents the maximum differential input that the circuit can "handle." It is possible to relate ΔV_{in1} to the overdrive voltage of M_1 and M_2 in equilibrium. For a zero differential input, $I_{D1} = I_{D2} = I_{SS}/2$, yielding

$$(V_{GS} - V_{TH})_{1,2} = \sqrt{\frac{I_{SS}}{\mu_n C_{ox}\dfrac{W}{L}}} \tag{4.15}$$

Thus, ΔV_{in1} is equal to $\sqrt{2}$ times the equilibrium overdrive. The point is that increasing ΔV_{in1} to make the circuit more linear inevitably increases the overdrive voltage of M_1 and M_2. For a given I_{SS}, this is accomplished only by reducing W/L and hence the transconductance of the transistors, trading small-signal gain for linearity. Alternatively, we can increase I_{SS}, but at the cost of power. (What happens to the gain if I_{SS} is increased but $I_{SS}R_D$ is kept constant due to headroom constraints?)

▶ Example 4.5

Due to a manufacturing defect, the differential signals applied to a differential pair have unequal dc levels (Fig. 4.15). If the peak swing, V_0, is small and the imbalance, V_{OS}, happens to be equal to $\Delta V_{in1}/2 = (1/2)\sqrt{2I_{SS}/(\mu_n C_{ox} W/L)}$, sketch the output voltage waveforms and determine the small-signal voltage gain.

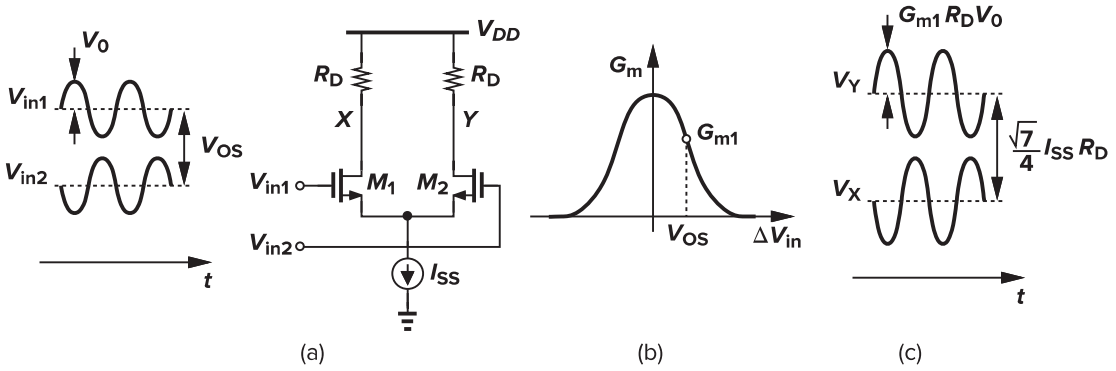


Figure 4.15

Solution

Let us first study the circuit with only dc inputs that differ by V_{OS}. The differential pair senses an imbalance of $V_{in1} - V_{in2} = V_{OS}$ and, from Eq. (4.10), generates

$$I_{D1} - I_{D2} = \frac{\sqrt{7}}{4} I_{SS} \tag{4.16}$$

That is, $I_{D1} \approx 0.83 I_{SS}$, $I_{D2} \approx 0.17 I_{SS}$ and $V_X - V_Y = -(\sqrt{7}/4) I_{SS} R_D$.

Now, we recognize from Fig. 4.15(b) that the input dc imbalance biases the transistors away from the highest transconductance, yielding from Eq. (4.11)

$$G_{m1} = \frac{3}{\sqrt{14}} \sqrt{\mu_n C_{ox} \frac{W}{L} I_{SS}} \tag{4.17}$$

This value is about 20% less than that at equilibrium. The output waveforms are shown in Fig. 4.15(c). ◀

Small-Signal Analysis We now study the small-signal behavior of differential pairs. As depicted in Fig. 4.16, we apply small signals V_{in1} and V_{in2} and assume that M_1 and M_2 are saturated. What is the differential voltage gain, $(V_{out1} - V_{out2})/(V_{in1} - V_{in2})$? Recall from Eq. (4.12) that this quantity equals $\sqrt{\mu_n C_{ox} I_{SS} W/L} R_D$. Since in the vicinity of equilibrium, each transistor carries approximately $I_{SS}/2$, this expression reduces to $g_m R_D$, where g_m denotes the transconductance of M_1 and M_2. To arrive at the same result by small-signal analysis, we employ two different methods, each providing insight into the circuit's operation. We assume that $R_{D1} = R_{D2} = R_D$.

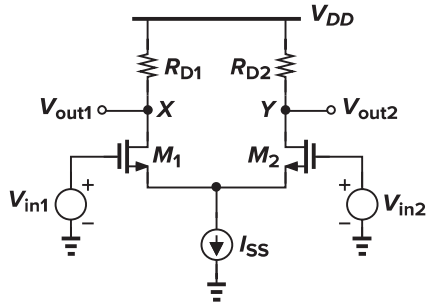


Figure 4.16 Differential pair with small-signal inputs.

Method I The circuit of Fig. 4.16 is driven by two independent signals. Thus, the output can be computed by superposition. (The voltages in this section are small-signal quantities.)

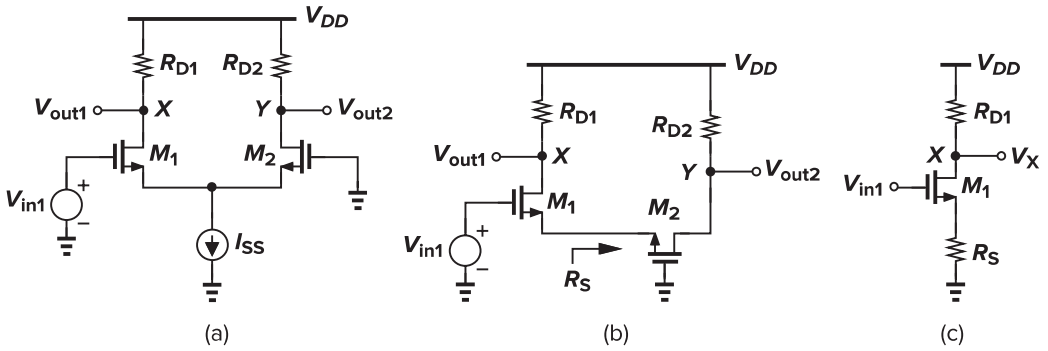


Figure 4.17 (a) Differential pair sensing one input signal; (b) circuit of (a) viewed as a CS stage degenerated by M_2; (c) equivalent circuit of (b).

Let us set V_{in2} to zero and find the effect of V_{in1} at X and Y [Fig. 4.17(a)]. To obtain V_X, we note that M_1 forms a common-source stage with a degeneration resistance equal to the impedance seen looking into the source of M_2 [Fig. 4.17(b)]. Neglecting channel-length modulation and body effect, we have $R_S = 1/g_{m2}$ [Fig. 4.17(c)] and

$$\frac{V_X}{V_{in1}} = \frac{-R_D}{\dfrac{1}{g_{m1}} + \dfrac{1}{g_{m2}}} \tag{4.18}$$

To calculate V_Y, we note that M_1 drives M_2 as a source follower and replace V_{in1} and M_1 by a Thevenin equivalent (Fig. 4.18): the Thevenin voltage $V_T = V_{in1}$ and the resistance $R_T = 1/g_{m1}$. Here, M_2 operates as a common-gate stage, exhibiting a gain equal to

$$\frac{V_Y}{V_{in1}} = \frac{R_D}{\dfrac{1}{g_{m2}} + \dfrac{1}{g_{m1}}} \tag{4.19}$$

It follows from (4.18) and (4.19) that the overall voltage gain for V_{in1} is

$$(V_X - V_Y)|_{\text{Due to } Vin1} = \frac{-2R_D}{\dfrac{1}{g_{m1}} + \dfrac{1}{g_{m2}}} V_{in1} \tag{4.20}$$

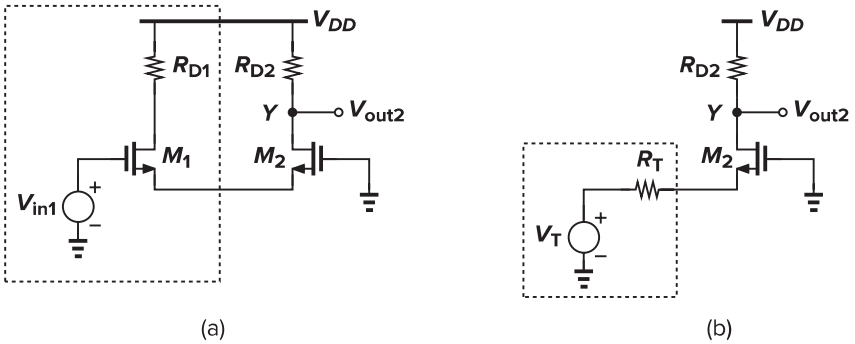


Figure 4.18 Replacing M_1 by a Thevenin equivalent.

which, for $g_{m1} = g_{m2} = g_m$, reduces to

$$(V_X - V_Y)|_{\text{Due to } Vin1} = -g_m R_D V_{in1} \tag{4.21}$$

By virtue of symmetry, the effect of V_{in2} at X and Y is identical to that of V_{in1} except for a change in the polarities:

$$(V_X - V_Y)|_{\text{Due to } Vin2} = g_m R_D V_{in2} \tag{4.22}$$

Adding the two sides of (4.21) and (4.22) to perform superposition, we have

$$\frac{(V_X - V_Y)_{tot}}{V_{in1} - V_{in2}} = -g_m R_D \tag{4.23}$$

Comparison of (4.21), (4.22), and (4.23) indicates that the magnitude of the differential gain is equal to $g_m R_D$ regardless of how the inputs are applied: in Figs. 4.17 and 4.18, the input is applied to only one side, whereas in Fig. 4.16 the input is the *difference* between two sources. It is also important to recognize that if the output is single-ended, i.e., it is sensed between X or Y and ground, the gain is halved.

▶ Example 4.6

Due to a manufacturing error, in the circuit of Fig. 4.19, M_2 is twice as wide as M_1. Calculate the small-signal gain if the dc levels of V_{in1} and V_{in2} are equal.

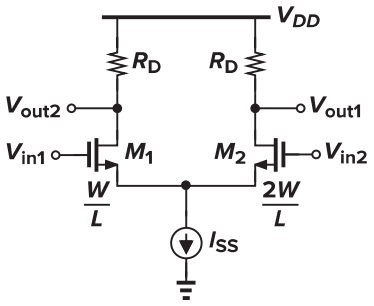


Figure 4.19

Solution

If the gates of M_1 and M_2 are at the same dc potential, then $V_{GS1} = V_{GS2}$ and $I_{D2} = 2I_{D1} = 2I_{SS}/3$. Thus, $g_{m1} = \sqrt{2\mu_n C_{ox}(W/L)I_{SS}/3}$ and $g_{m2} = \sqrt{2\mu_n C_{ox}(2W/L)(2I_{SS})/3} = 2g_{m1}$. Following the same procedure as above, the reader can show that

$$|A_v| = \frac{2R_D}{\dfrac{1}{g_{m1}} + \dfrac{1}{2g_{m1}}} \tag{4.24}$$

$$= \frac{4}{3} g_{m1} R_D \tag{4.25}$$

Note that, for a given I_{SS}, this value is lower than the gain of a symmetric differential pair [Eq. (4.23)] because g_{m1} is smaller. The reader can show that the characteristics of Fig. 4.13 are shifted horizontally, and hence the circuit exhibits an "offset." We utilize this idea in Chapter 14 to linearize differential pairs.

◀

How does the transconductance of a differential pair compare with that of a common-source stage? For a given *total* bias current, the value of g_m in (4.23) is $1/\sqrt{2}$ times that of a single transistor biased at I_{SS} with the same dimensions. Thus, the total G_m is proportionally less.

Method II If a fully-symmetric differential pair senses differential inputs (i.e., the two inputs change by equal and opposite amounts from the equilibrium condition), then the concept of "half circuit" can be applied. We first prove a lemma.

Lemma Consider the symmetric circuit shown in Fig. 4.20(a), where D_1 and D_2 represent any three-terminal active device. Suppose V_{in1} and V_{in2} change differentially, the former from V_0 to $V_0 + \Delta V_{in}$ and the latter from V_0 to $V_0 - \Delta V_{in}$ [Fig. 4.20(b)]. Then, if the circuit remains linear, V_P does not change. Assume $\lambda = 0$.

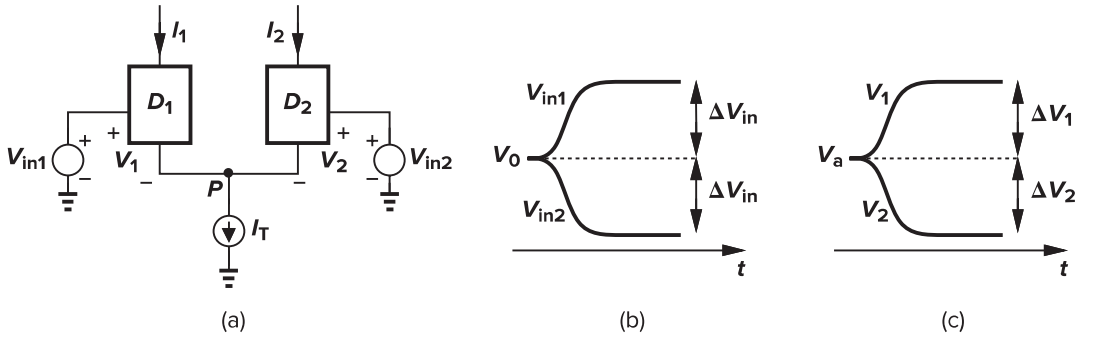


Figure 4.20 Illustration of why node P is a virtual ground.

Proof. The lemma can be proved by invoking symmetry. As long as the operation remains linear, so that the difference between the bias currents of D_1 and D_2 is negligible, the circuit is symmetric. Thus, V_P cannot "favor" the change at one input and "ignore" the other.

From another point of view, the effect of D_1 and D_2 at node P can be represented by Thevenin equivalents (Fig. 4.21). If V_{T1} and V_{T2} change by equal and opposite amounts and R_{T1} and R_{T2} are equal, then V_P remains constant. We emphasize that this is valid if the changes are small enough that we can assume $R_{T1} = R_{T2}$ (e.g., $1/g_{m1} = 1/g_{m2}$).[6] This perspective suggests the lemma's validity even if the tail current source is not ideal. ❑

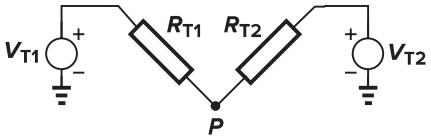


Figure 4.21 Replacing each half of a differential pair by a Thevenin equivalent.

We now offer a more formal proof. Let us assume that V_1 and V_2 have an equilibrium value of V_a and change by ΔV_1 and ΔV_2, respectively [Fig. 4.20(c)]. The output currents therefore change by $g_m \Delta V_1$ and $g_m \Delta V_2$. Since $I_1 + I_2 = I_T$, we have $g_m \Delta V_1 + g_m \Delta V_2 = 0$, i.e., $\Delta V_1 = -\Delta V_2$. We also know that $V_{in1} - V_1 = V_{in2} - V_2$, and hence $V_0 + \Delta V_{in} - (V_a + \Delta V_1) = V_0 - \Delta V_{in} - (V_a + \Delta V_2)$. Consequently, $2\Delta V_{in} = \Delta V_1 - \Delta V_2 = 2\Delta V_1$. In other words, if V_{in1} and V_{in2} change by $+\Delta V_{in}$ and $-\Delta V_{in}$, respectively, then V_1 and V_2 change by the same values, i.e., a differential change in the inputs is simply "absorbed" by V_1 and V_2. In fact, since $V_P = V_{in1} - V_1$, and since V_1 exhibits the same change as V_{in1}, V_P does not change.

[6] It is also possible to derive an expression for the large-signal behavior of V_P and prove that for small $V_{in1} - V_{in2}$, V_P remains constant. We defer this calculation to Chapter 15.

The above lemma greatly simplifies the small-signal analysis of differential amplifiers. As shown in Fig. 4.22, since V_P experiences no change, node P can be considered "ac ground" (or a "virtual ground"), and the circuit can be decomposed into two separate halves. We say that we have applied the "half-circuit concept" [1]. We can write $V_X/V_{in1} = -g_m R_D$ and $V_Y/(-V_{in1}) = -g_m R_D$, where V_{in1} and $-V_{in1}$ denote the voltage *change* on each side. Thus, $(V_X - V_Y)/(2V_{in1}) = -g_m R_D$.

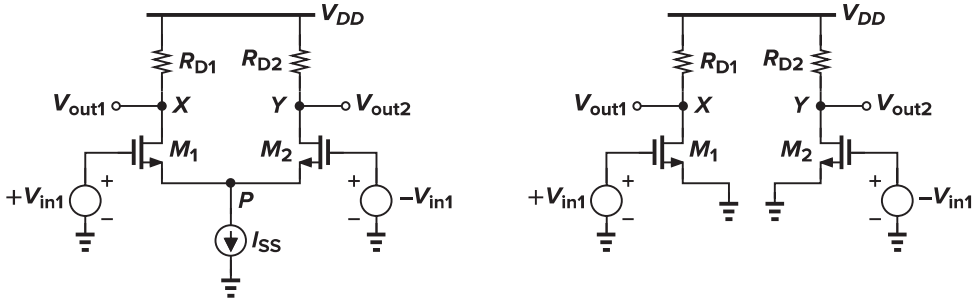


Figure 4.22 Application of the half-circuit concept.

▶ **Example 4.7**

Calculate the differential gain of the circuit of Fig. 4.22(a) if $\lambda \neq 0$.

Solution

Applying the half-circuit concept as illustrated in Fig. 4.23, we have $V_X/V_{in1} = -g_m(R_D \| r_{O1})$ and $V_Y/(-V_{in1}) = -g_m(R_D \| r_{O2})$, thus arriving at $(V_X - V_Y)/(2V_{in1}) = -g_m(R_D \| r_O)$, where $r_O = r_{O1} = r_{O2}$. Note that Method I would require lengthy calculations here.

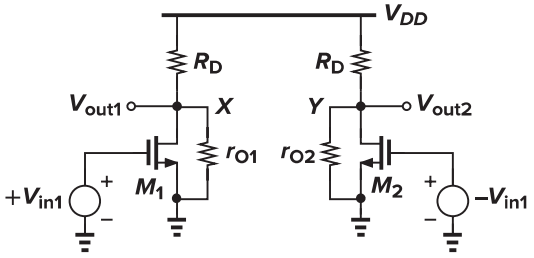


Figure 4.23

◀

The half-circuit concept provides a powerful technique for analyzing symmetric differential pairs with fully differential inputs. But what happens if the two inputs are not fully differential [Fig. 4.24(a)]? As depicted in Figs. 4.24(b) and (c), the two inputs V_{in1} and V_{in2} can be viewed as

$$V_{in1} = \frac{V_{in1} - V_{in2}}{2} + \frac{V_{in1} + V_{in2}}{2} \tag{4.26}$$

$$V_{in2} = \frac{V_{in2} - V_{in1}}{2} + \frac{V_{in1} + V_{in2}}{2} \tag{4.27}$$

Since the second term is common to both inputs, we obtain the equivalent circuit in Fig. 4.24(d), recognizing that the circuit senses a combination of a differential input and a common-mode variation. Therefore, as illustrated in Fig. 4.25, the effect of each type of input can be computed by superposition, with the half-circuit concept applied to the differential-mode operation. We deal with CM analysis in Sec. 4.3.

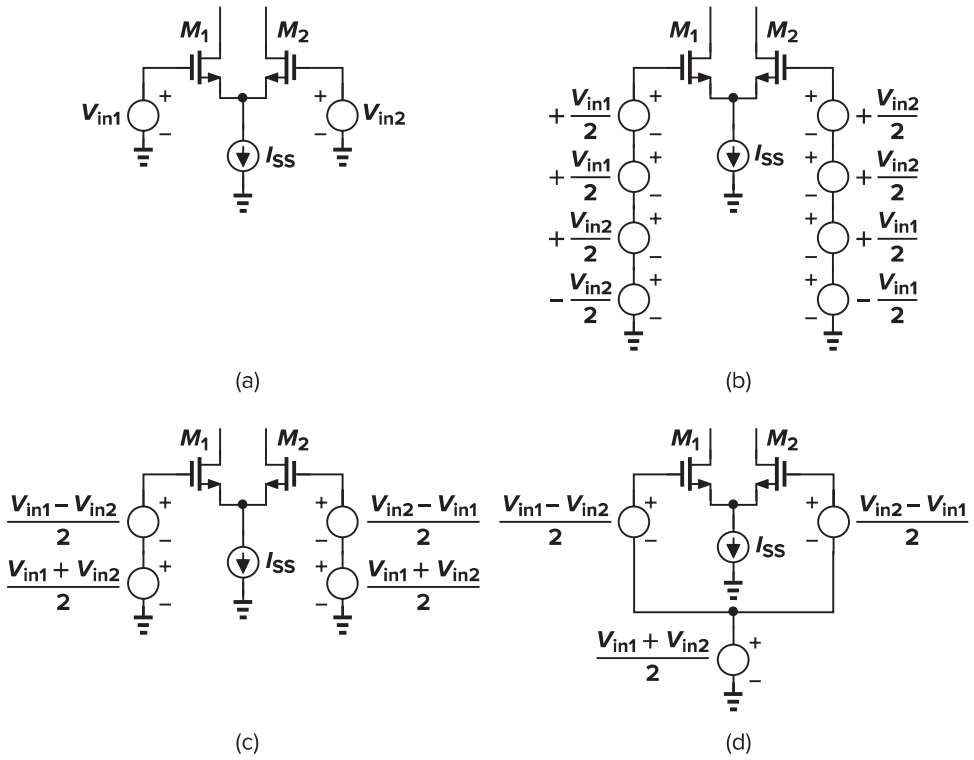


Figure 4.24 Conversion of arbitrary inputs to differential and common-mode components.

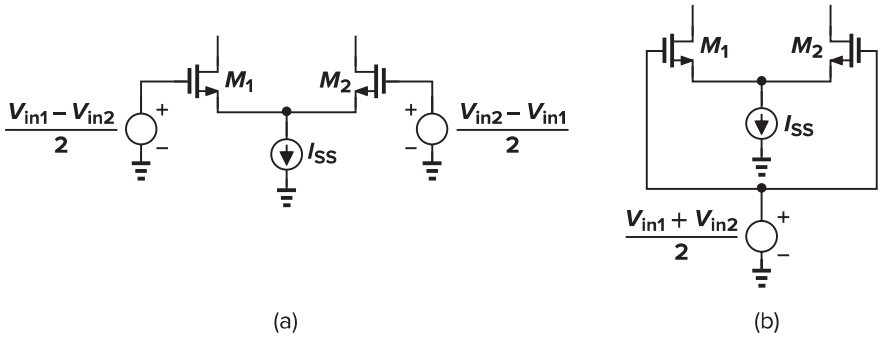


Figure 4.25 Superposition for (a) differential and (b) common-mode signals.

▶ Example 4.8

In the circuit of Fig. 4.22(a), calculate V_X and V_Y if $V_{in1} \neq -V_{in2}$ and $\lambda \neq 0$.

Solution

For differential-mode operation, we have from Fig. 4.26(a)

$$V_X = -g_m(R_D \| r_{O1})\frac{V_{in1} - V_{in2}}{2} \tag{4.28}$$

$$V_Y = -g_m(R_D \| r_{O2})\frac{V_{in2} - V_{in1}}{2} \tag{4.29}$$

That is,

$$V_X - V_Y = -g_m(R_D \| r_O)(V_{in1} - V_{in2}) \tag{4.30}$$

which is to be expected.

For common-mode operation, the circuit reduces to that in Fig. 4.26(b). How much do V_X and V_Y change as $V_{in,CM}$ changes? If the circuit is fully symmetric and I_{SS} an ideal current source, the currents drawn by M_1 and M_2 from R_{D1} and R_{D2} are exactly equal to $I_{SS}/2$ and independent of $V_{in,CM}$. Thus, V_X and V_Y remain equal to $V_{DD} - R_D(I_{SS}/2)$ and experience no change as $V_{in,CM}$ varies. Interestingly, the circuit simply amplifies the difference between V_{in1} and V_{in2} while eliminating the effect of $V_{in,CM}$.

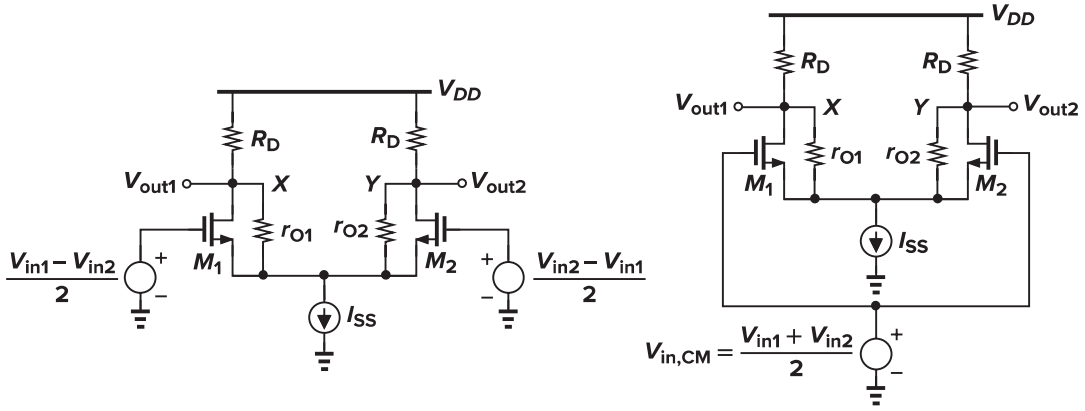


Figure 4.26

4.2.3 Degenerated Differential Pair

As with a simple common-source stage, a differential pair can incorporate resistive degeneration to improve its linearity. Shown in Fig. 4.27(a), such a topology softens the nonlinear behavior of M_1 and M_2 by R_{S1} and R_{S2}. This can be seen from the input-output characteristics of Fig. 4.27(b), where, due to degeneration, the differential voltage necessary to turn off one side increases in magnitude. We can

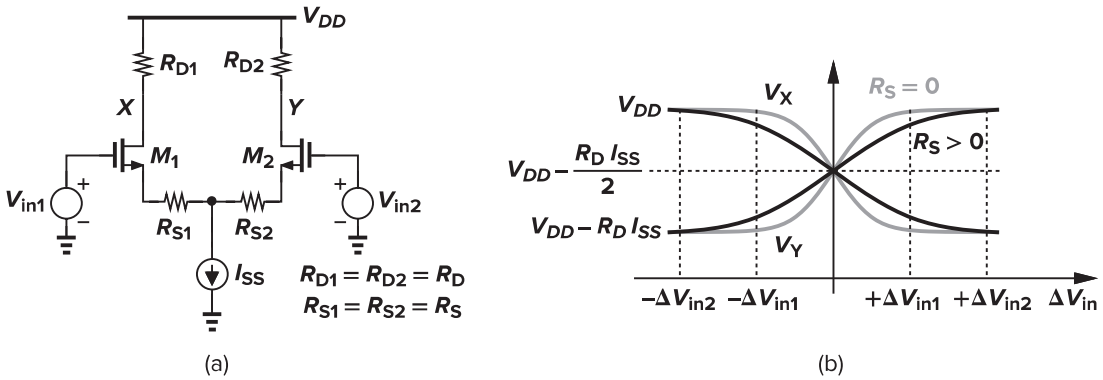


Figure 4.27 (a) Degenerated differential pair, and (b) characteristics with and without degeneration.

readily prove this point. Suppose that at $V_{in1} - V_{in2} = \Delta V_{in2}$, M_2 turns off and $I_{D1} = I_{SS}$. We then have $V_{GS2} = V_{TH}$, and hence

$$V_{in1} - V_{GS1} - R_S I_{SS} = V_{in2} - V_{TH} \tag{4.31}$$

which yields

$$V_{in1} - V_{in2} = V_{GS1} - V_{TH} + R_S I_{SS} \tag{4.32}$$

$$= \sqrt{\frac{2I_{SS}}{\mu_n C_{ox} \dfrac{W}{L}}} + R_S I_{SS} \tag{4.33}$$

We recognize the first term on the right-hand side as ΔV_{in1} (the input difference necessary for turning off M_2 if $R_S = 0$). It follows that

$$\Delta V_{in2} - \Delta V_{in1} = R_S I_{SS} \tag{4.34}$$

suggesting that the linear input range is widened by approximately $\pm R_S I_{SS}$.

The small-signal voltage gain of the degenerated differential pair can be obtained by applying the half-circuit concept. The half circuit is simply a degenerated CS stage, exhibiting a gain of

$$|A_v| = \frac{R_D}{\dfrac{1}{g_m} + R_S} \tag{4.35}$$

if $\lambda = \gamma = 0$. The circuit thus trades gain for linearity—as is also observed from the slopes of the characteristics in Fig. 4.27(b). Note that A_v is less sensitive to g_m variations in this case.

In addition to reducing the gain, the degeneration resistors in Fig. 4.27(a) also consume voltage headroom. In the equilibrium condition, each resistor sustains a voltage drop of $R_S I_{SS}/2$, as if the tail current source itself required this much more headroom. The input common-mode level must therefore be higher by this amount, and so must be the minimum voltage at X or Y. In other words, the maximum allowable *differential* output swing is reduced by $R_S I_{SS}$. This issue can be resolved as shown in Fig. 4.28, where the tail current source is split in half, with each half directly tied to a source. In equilibrium, no current flows through the degeneration resistance, and hence no headroom is sacrificed.[7] Other methods of linearizing differential pairs are described in Chapter 14.

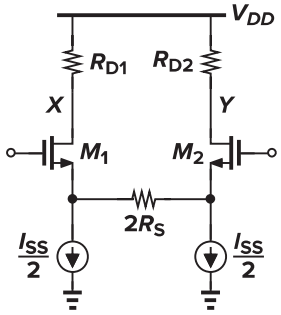


Figure 4.28 Degenerated differential pair with split tail current source.

[7] But, as explained later in the book, the two tail current sources do contribute differential noise and offset in this case.

4.3 ■ Common-Mode Response

An important attribute of differential amplifiers is their ability to suppress the effect of common-mode perturbations. Example 4.8 portrays an idealized case of common-mode response. In reality, neither is the circuit fully symmetric nor does the current source exhibit an infinite output impedance. As a result, a fraction of the input CM variation appears at the output.

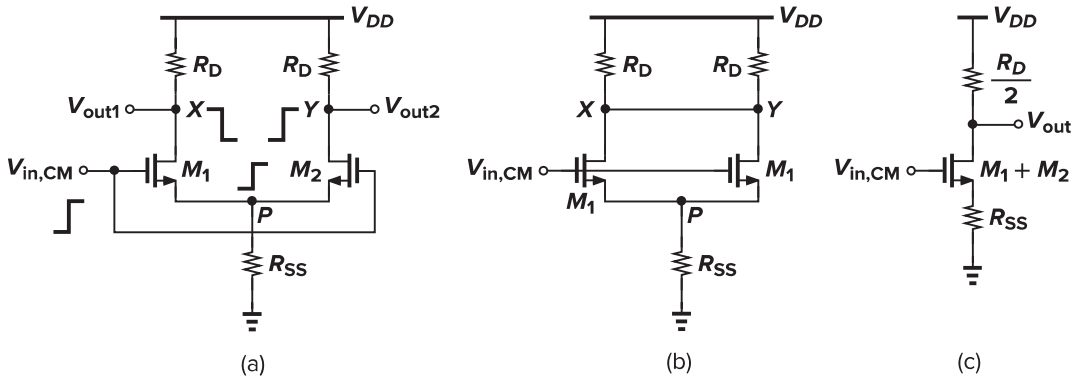


Figure 4.29 (a) Differential pair sensing CM input; (b) simplified version of (a); (c) equivalent circuit of (b).

We first assume that the circuit is symmetric, but the current source has a finite output impedance, R_{SS} [Fig. 4.29(a)]. As $V_{in,CM}$ changes, so does V_P, thereby increasing the drain currents of M_1 and M_2 and lowering both V_X and V_Y. Owing to symmetry, V_X remains equal to V_Y and, as depicted in Fig. 4.29(b), the two nodes can be shorted together. Since M_1 and M_2 are now "in parallel," i.e., they share all of their respective terminals, the circuit can be reduced to that in Fig. 4.29(c). Note that the composite device, $M_1 + M_2$, has twice the width and the bias current of each of M_1 and M_2 and, therefore, twice their transconductance. The "common-mode gain" of the circuit is thus equal to

$$A_{v,CM} = \frac{V_{out}}{V_{in,CM}} \tag{4.36}$$

$$= -\frac{R_D/2}{1/(2g_m) + R_{SS}} \tag{4.37}$$

where g_m denotes the transconductance of each of M_1 and M_2 and $\lambda = \gamma = 0$.

What is the significance of this calculation? In a symmetric circuit, input CM variations disturb the bias points, altering the small-signal gain and possibly limiting the output voltage swings. This can be illustrated by an example.

▶ Example 4.9

The circuit of Fig. 4.30 uses a resistor rather than a current source to define a tail current of 1 mA. Assume that $(W/L)_{1,2} = 25/0.5$, $\mu_n C_{ox} = 50\ \mu\text{A/V}^2$, $V_{TH} = 0.6$ V, $\lambda = \gamma = 0$, and $V_{DD} = 3$ V.

(a) What is the required input CM voltage for which R_{SS} sustains 0.5 V?

(b) Calculate R_D for a differential gain of 5.

(c) What happens at the output if the input CM level is 50 mV higher than the value calculated in (a)?

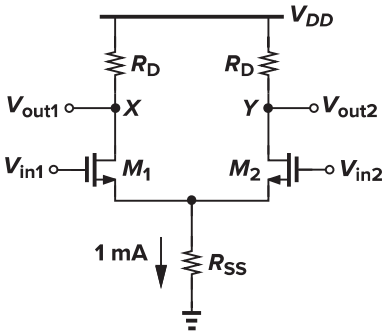


Figure 4.30

Solution

(a) Since $I_{D1} = I_{D2} = 0.5$ mA, we have

$$V_{GS1} = V_{GS2} = \sqrt{\frac{2I_{D1}}{\mu_n C_{ox}\dfrac{W}{L}}} + V_{TH} \tag{4.38}$$

$$= 1.23 \text{ V} \tag{4.39}$$

Thus, $V_{in,CM} = V_{GS1} + 0.5$ V $= 1.73$ V. Note that $R_{SS} = 500\ \Omega$.

(b) The transconductance of each device is $g_m = \sqrt{2\mu_n C_{ox}(W/L)I_{D1}} = 1/(632\ \Omega)$, requiring $R_D = 3.16$ kΩ for a gain of 5.

Note that the output bias level is equal to $V_{DD} - I_{D1}R_D = 1.42$ V. Since $V_{in,CM} = 1.73$ V and $V_{TH} = 0.6$ V, the transistors are 290 mV away from the triode region.

(c) If $V_{in,CM}$ increases by 50 mV, the equivalent circuit of Fig. 4.29(c) suggests that V_X and V_Y drop by

$$|\Delta V_{X,Y}| = \Delta V_{in,CM}\frac{R_D/2}{R_{SS} + 1/(2g_m)} \tag{4.40}$$

$$= 50 \text{ mV} \times 1.94 \tag{4.41}$$

$$= 96.8 \text{ mV} \tag{4.42}$$

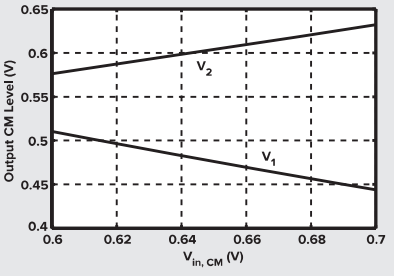
Now, M_1 and M_2 are only 143 mV away from the triode region because the input CM level has increased by 50 mV and the output CM level has decreased by 96.8 mV. ◀

The foregoing discussion indicates that the finite output impedance of the tail current source results in some common-mode gain in a symmetric differential pair. Nonetheless, this is usually a minor concern. More troublesome is the variation of the *differential* output as a result of a change in $V_{in,CM}$, an effect that occurs because in reality the circuit is not fully symmetric, i.e., the two sides suffer from slight mismatches during manufacturing. For example, in Fig. 4.29(a), R_{D1} may not be exactly equal to R_{D2}.

We now study the effect of input common-mode variations if the circuit is asymmetric and the tail current source suffers from a finite output impedance. Suppose, as shown in Fig. 4.31, $R_{D1} = R_D$ and $R_{D2} = R_D + \Delta R_D$, where ΔR_D denotes a small mismatch and the circuit is otherwise symmetric. Assume that $\lambda = \gamma = 0$ for M_1 and M_2. What happens to V_X and V_Y as $V_{in,CM}$ increases? We recognize that M_1 and M_2 operate as one source follower,

Nanometer Design Notes

As a result of the low output impedance of tail current sources in nanometer technologies, a CM level change can "propagate." Plotted below are the output CM levels of two cascaded differential pairs as the main input CM level, $V_{in,CM}$, increases, revealing a drop in the first and a rise in the second.



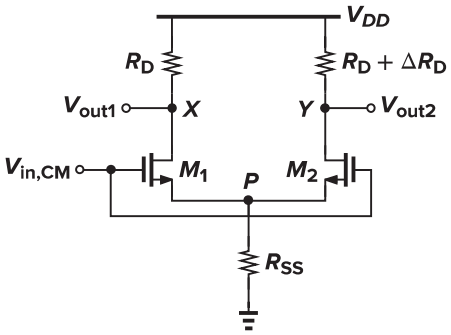


Figure 4.31 Common-mode response in the presence of resistor mismatch.

raising V_P by

$$\Delta V_P = \frac{R_{SS}}{R_{SS} + \dfrac{1}{2g_m}} \Delta V_{in,CM} \tag{4.43}$$

Since M_1 and M_2 are identical, I_{D1} and I_{D2} increase by $[g_m/(1 + 2g_m R_{SS})]\Delta V_{in,CM}$, but V_X and V_Y change by different amounts:

$$\Delta V_X = -\Delta V_{in,CM} \frac{g_m}{1 + 2g_m R_{SS}} R_D \tag{4.44}$$

$$\Delta V_Y = -\Delta V_{in,CM} \frac{g_m}{1 + 2g_m R_{SS}} (R_D + \Delta R_D) \tag{4.45}$$

Thus, a common-mode change at the input introduces a *differential* component at the output. We say that the circuit exhibits common-mode to differential conversion. This is a critical problem because if the input of a differential pair includes both a differential signal and common-mode noise, the circuit corrupts the amplified differential signal by the input CM change. The effect is illustrated in Fig. 4.32.

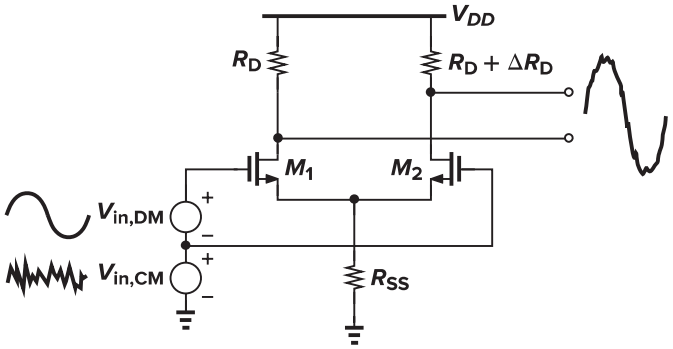


Figure 4.32 Effect of CM noise in the presence of resistor mismatch.

In summary, the common-mode response of differential pairs depends on the output impedance of the tail current source and asymmetries in the circuit, manifesting itself through two effects: variation of the output CM level (in the absence of mismatches) and conversion of input common-mode variations to differential components at the output. In analog circuits, the latter effect is much more severe than the

former. For this reason, the common-mode response should usually be studied with mismatches taken into account.

How significant is common-mode to differential conversion? We make two observations. First, as the *frequency* of the CM disturbance increases, the total capacitance shunting the tail current source introduces larger tail current variations. Thus, even if the output *resistance* of the current source is high, common-mode to differential conversion becomes significant at high frequencies. Shown in Fig. 4.33, this capacitance arises from the parasitics of the current source itself as well as the source-bulk junctions of M_1 and M_2. Second, the asymmetry in the circuit stems from both the load resistors and the input transistors, the latter contributing a typically much greater mismatch.

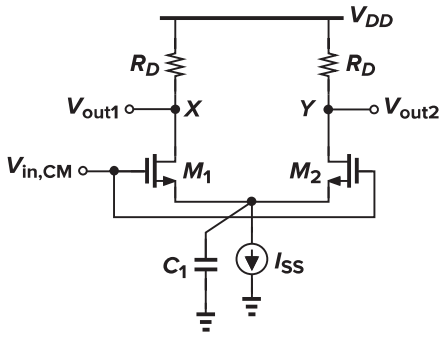


Figure 4.33 CM response with finite tail capacitance.

Let us study the asymmetry resulting from mismatches between M_1 and M_2 in Fig. 4.34(a). Owing to dimension and threshold voltage mismatches, the two transistors carry slightly different currents and exhibit unequal transconductances. We assume that $\lambda = \gamma = 0$. To calculate the small-signal gain from $V_{in,CM}$ to X and Y, we use the equivalent circuit in Fig. 4.34(b), writing $I_{D1} = g_{m1}(V_{in,CM} - V_P)$ and $I_{D2} = g_{m2}(V_{in,CM} - V_P)$. Since $(I_{D1} + I_{D2})R_{SS} = V_P$,

$$(g_{m1} + g_{m2})(V_{in,CM} - V_P)R_{SS} = V_P \tag{4.46}$$

and

$$V_P = \frac{(g_{m1} + g_{m2})R_{SS}}{(g_{m1} + g_{m2})R_{SS} + 1} V_{in,CM} \tag{4.47}$$

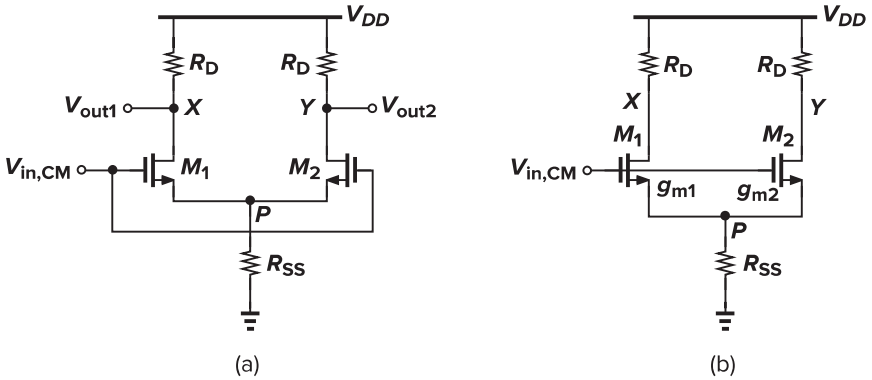


Figure 4.34 (a) Differential pair sensing CM input; (b) equivalent circuit of (a).

We now obtain the output voltages as

$$V_X = -g_{m1}(V_{in,CM} - V_P)R_D \tag{4.48}$$

$$= \frac{-g_{m1}}{(g_{m1} + g_{m2})R_{SS} + 1} R_D V_{in,CM} \tag{4.49}$$

and

$$V_Y = -g_{m2}(V_{in,CM} - V_P)R_D \tag{4.50}$$

$$= \frac{-g_{m2}}{(g_{m1} + g_{m2})R_{SS} + 1} R_D V_{in,CM} \tag{4.51}$$

The differential component at the output is therefore given by

$$V_X - V_Y = -\frac{g_{m1} - g_{m2}}{(g_{m1} + g_{m2})R_{SS} + 1} R_D V_{in,CM} \tag{4.52}$$

In other words, the circuit converts input CM variations to a differential error by a factor equal to

$$A_{CM-DM} = -\frac{\Delta g_m R_D}{(g_{m1} + g_{m2})R_{SS} + 1} \tag{4.53}$$

where A_{CM-DM} denotes common-mode to differential-mode conversion and $\Delta g_m = g_{m1} - g_{m2}$.

▶ Example 4.10

Two differential pairs are cascaded as shown in Fig. 4.35. Transistors M_3 and M_4 suffer from a g_m mismatch of Δg_m, the total parasitic capacitance at node P is represented by C_P, and the circuit is otherwise symmetric. What fraction of the supply noise appears as a differential component at the output? Assume that $\lambda = \gamma = 0$.

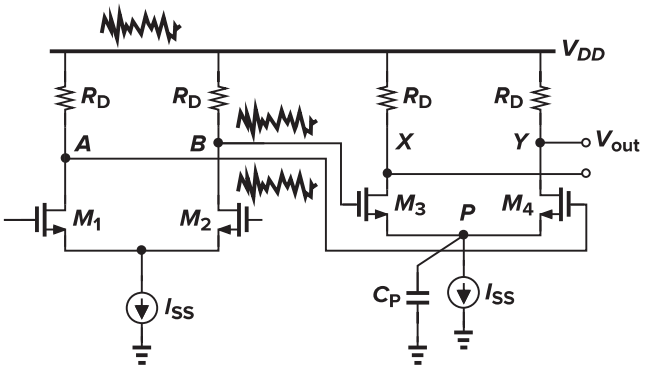


Figure 4.35

Solution

Neglecting the capacitance at nodes A and B, we note that the supply noise appears at these nodes with no attenuation. Substituting $1/(C_P s)$ for R_{SS} in (4.53) and taking the magnitude, we have

$$|A_{CM-DM}| = \frac{\Delta g_m R_D}{\sqrt{1 + (g_{m3} + g_{m4})^2 \left| \frac{1}{C_P \omega} \right|^2}} \tag{4.54}$$

The key point is that the effect becomes more noticeable as the supply noise frequency, ω, increases. ◀

For a meaningful comparison of differential circuits, the undesirable differential component produced by CM variations must be normalized to the wanted differential output resulting from amplification. We define the "common-mode rejection ratio" (CMRR) as the desired gain divided by the undesired gain:

$$\text{CMRR} = \left| \frac{A_{DM}}{A_{CM-DM}} \right| \tag{4.55}$$

If only g_m mismatch is considered, the reader can show from the analysis of Fig. 4.17 that

$$|A_{DM}| = \frac{R_D}{2} \frac{g_{m1} + g_{m2} + 4g_{m1}g_{m2}R_{SS}}{1 + (g_{m1} + g_{m2})R_{SS}} \tag{4.56}$$

and hence

$$\text{CMRR} = \frac{g_{m1} + g_{m2} + 4g_{m1}g_{m2}R_{SS}}{2\Delta g_m} \tag{4.57}$$

$$\approx \frac{g_m}{\Delta g_m}(1 + 2g_m R_{SS}) \tag{4.58}$$

where g_m denotes the mean value, that is, $g_m = (g_{m1} + g_{m2})/2$. In practice, all mismatches must be taken into account. Note that $2g_m R_{SS} \gg 1$, and hence CMRR $\approx 2g_m^2 R_{SS}/\Delta g_m$.

▶ **Example 4.11**

Our studies suggest that an ideal tail current source guarantees infinite CM rejection. Is this always true?

Solution

Interestingly, it is not. If the two transistors exhibit body-effect mismatch, then the circuit still converts an input CM change to a differential output component even if the tail impedance is infinite. As illustrated in Fig. 4.36, a change in $V_{in,CM}$ produces a change in V_P, and hence in V_{BS} of both transistors. If $g_{mb1} \neq g_{mb2}$, the change in $I_{D1}(= g_{mb1}V_{BS1})$ is not equal to that in I_{D2}, yielding a differential change at the output.

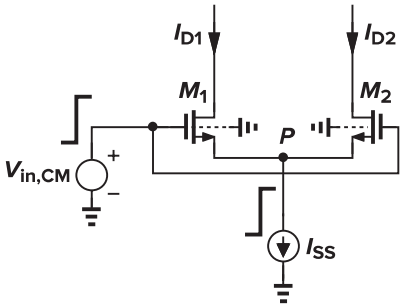


Figure 4.36

◀

4.4 ■ Differential Pair with MOS Loads

The load of a differential pair need not be implemented by linear resistors. As with the common-source stages studied in Chapter 3, differential pairs can employ diode-connected or current-source loads (Fig. 4.37). The small-signal differential gain can be derived using the half-circuit concept. For Fig. 4.37(a),

$$A_v = -g_{mN}\left(g_{mP}^{-1} \| r_{ON} \| r_{OP}\right) \tag{4.59}$$

$$\approx -\frac{g_{mN}}{g_{mP}} \tag{4.60}$$

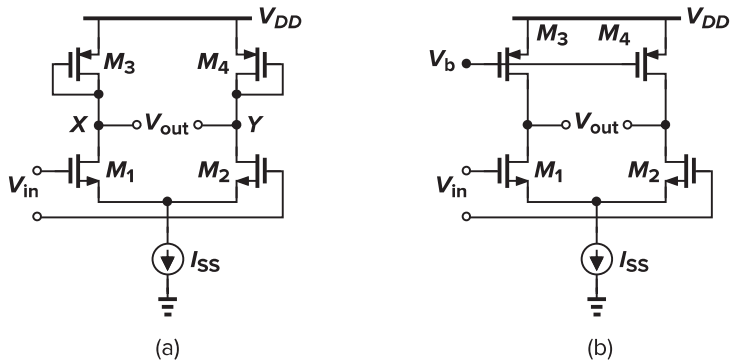


Figure 4.37 Differential pair with (a) diode-connected and (b) current-source loads.

where the subscripts N and P denote NMOS and PMOS, respectively. Expressing g_{mN} and g_{mP} in terms of device dimensions, we have

$$A_v \approx -\sqrt{\frac{\mu_n (W/L)_N}{\mu_p (W/L)_P}} \tag{4.61}$$

For Fig. 4.37(b), we have

$$A_v = -g_{mN}(r_{ON} \| r_{OP}) \tag{4.62}$$

▶ Example 4.12

It is possible to obviate the need for V_b in the circuit of Fig. 4.37(b) as shown in Fig. 4.38(a), where R_1 and $R_2 (= R_1)$ are relatively large. In the absence of signals, $V_X = V_Y = V_N = V_{DD} - |V_{GS3,4}|$. That is, M_3 and M_4 are "self-biased." Determine the differential voltage gain of this topology.

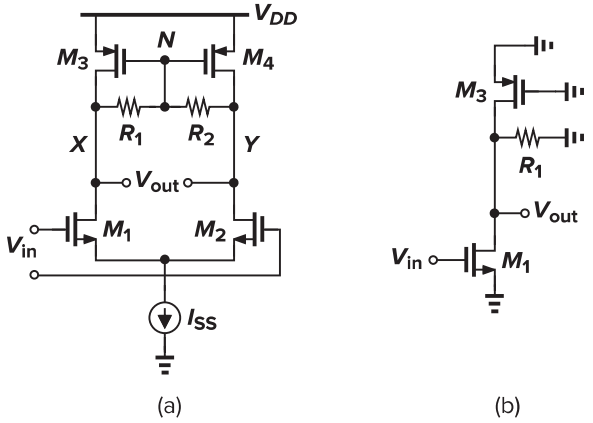


Figure 4.38

Solution

For differential outputs, V_N does not change (why?) and can be considered ac ground. Shown in Fig. 4.38(b), the half-circuit yields

$$|A_v| = g_{m1}(r_{O1} \| R_1 \| r_{O3}) \tag{4.63}$$

If the resistors are much greater than $r_{O1} || r_{O3}$, then they negligibly reduce the gain. ◀

In the circuit of Fig. 4.37(a), the diode-connected loads consume voltage headroom, thus creating a trade-off between the output voltage swings, the voltage gain, and the input CM range. Recall from Eq. (3.37) that, for given bias current and input device dimensions, the circuit's gain and the PMOS overdrive voltage scale together. To achieve a higher gain, $(W/L)_P$ must decrease, thereby increasing $|V_{GSP} - V_{THP}|$ and lowering the CM level at nodes X and Y.

In order to alleviate the above difficulty, part of the bias currents of the input transistors can be provided by PMOS current sources. Illustrated in Fig. 4.39(a), the idea is to lower the g_m of the load devices by reducing their current rather than their aspect ratio. For example, if the "auxiliary" current sources, M_5 and M_6, carry 80% of the drain current of M_1 and M_2, the current through M_3 and M_4 is reduced by a factor of five. For a given $|V_{GSP} - V_{THP}|$, this translates to a fivefold reduction in the transconductance of M_3 and M_4 because the aspect ratio of the devices can be lowered by the same factor. Thus, the differential gain is now five times that of the case with no PMOS current sources (if $\lambda = 0$).

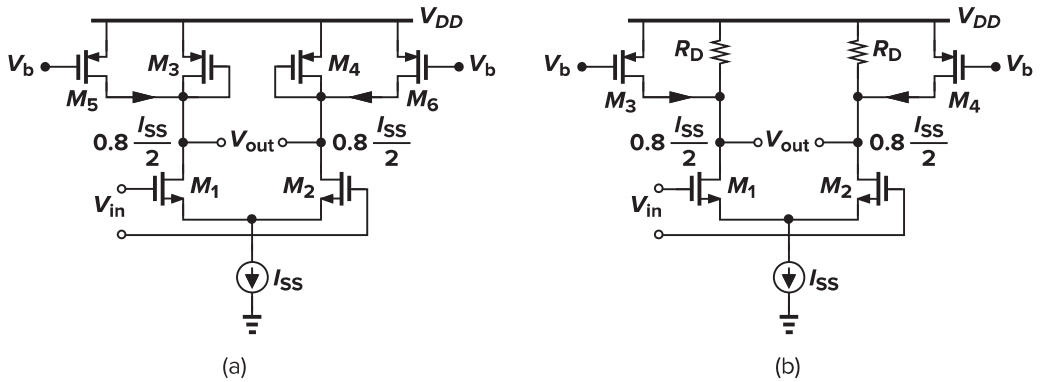


Figure 4.39 Addition of current sources to increase the voltage gain with (a) diode-connected loads and (b) resistive loads.

Since the voltage headroom consumed by diode-connected devices cannot be less than V_{TH} (if subthreshold conduction is neglected), the topology of Fig. 4.39(a) allows limited output voltage swings. We therefore prefer the alternative shown in Fig. 4.39(b), where the loads are realized by resistors—and the maximum voltage at each output node is equal to $V_{DD} - |V_{GS3,4} - V_{TH3,4}|$ rather than $V_{DD} - |V_{TH3,4}|$. For a given output CM level and 80% auxiliary currents, R_D can be five times as large, yielding a voltage gain of

$$|A_v| = g_{mN}(R_D||r_{ON}||r_{OP}) \tag{4.64}$$

If the PMOS devices are long (and, necessarily, wide), then $r_{OP} \gg r_{ON}$ and the gain is limited by $R_D||r_{ON}$. The circuit of Fig. 4.39(b) approaches that in Fig. 4.37(b) if $R_D \to \infty$, with the PMOS current sources providing *all* of the bias currents of M_1 and M_2.

The small-signal gain of the differential pair with current-source loads is relatively low—in the range of 5 to 10 in nanometer technologies. How do we increase the voltage gain? Borrowing ideas from the amplifiers in Chapter 3, we increase the output impedance of both the PMOS and the NMOS devices by cascoding, in essence creating a differential version of the cascode stage introduced in Chapter 3. The result is depicted in Fig. 4.40(a). To calculate the gain, we construct the half circuit of Fig. 4.40(b), which is similar to the cascode stage of Fig. 3.70. It follows that

$$|A_v| \approx g_{m1}[(g_{m3}r_{O3}r_{O1})||(g_{m5}r_{O5}r_{O7})] \tag{4.65}$$

Cascoding therefore increases the differential gain substantially, but at the cost of consuming more voltage headroom. We return to this circuit in Chapter 9.

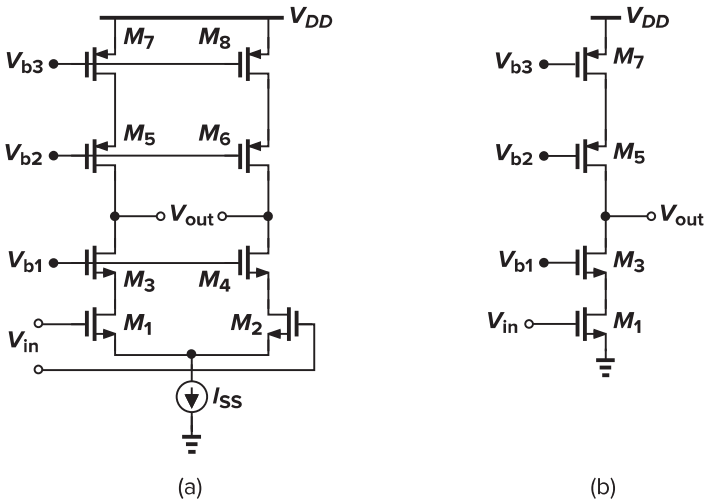


Figure 4.40 (a) Cascode differential pair; (b) half circuit of (a).

As a final note, we should mention that high-gain fully differential amplifiers require a means of defining the output common-mode level. For example, in Fig. 4.37(b), the output common-mode level is not well-defined, whereas in Fig. 4.37(a), diode-connected transistors define the output CM level as $V_{DD} - V_{GSP}$. We revisit this issue in Chapter 9.

4.5 ■ Gilbert Cell

Our study of differential pairs reveals two important aspects of their operation: (1) the small-signal gain of the circuit is a function of the tail current, and (2) the two transistors in a differential pair provide a simple means of steering the tail current to one of two destinations. By combining these two properties, we can develop a versatile building block.

Suppose we wish to construct a differential pair whose gain is varied by a control voltage. This can be accomplished as depicted in Fig. 4.41(a), where the control voltage defines the tail current and hence the gain. In this topology, $A_v = V_{out}/V_{in}$ varies from zero (if $I_{D3} = 0$) to a maximum value given by voltage headroom limitations and device dimensions. This circuit is a simple example of a "variable-gain amplifier" (VGA). VGAs find application in systems where the signal amplitude may experience large variations and hence requires inverse changes in the gain.

Now suppose we seek an amplifier whose gain can be continuously varied from a *negative* value to a *positive* value. Consider two differential pairs that amplify the input by opposite gains [Fig. 4.41(b)]. We now have $V_{out1}/V_{in} = -g_m R_D$ and $V_{out2}/V_{in} = +g_m R_D$, where g_m denotes the transconductance of each transistor in equilibrium. If I_1 and I_2 vary in opposite directions, so do $|V_{out1}/V_{in}|$ and $|V_{out2}/V_{in}|$.

But how should V_{out1} and V_{out2} be combined into a single output? As illustrated in Fig. 4.42(a), the two voltages can be summed, producing $V_{out} = V_{out1} + V_{out2} = A_1 V_{in} + A_2 V_{in}$, where A_1 and A_2 are controlled by V_{cont1} and V_{cont2}, respectively. The actual implementation is in fact quite simple: since $V_{out1} = R_D I_{D1} - R_D I_{D2}$ and $V_{out2} = R_D I_{D4} - R_D I_{D3}$, we have $V_{out1} + V_{out2} = R_D(I_{D1} + I_{D4}) - R_D(I_{D2} + I_{D3})$. Thus, rather than add V_{out1} and V_{out2}, we simply short the corresponding drain terminals to sum the currents and subsequently generate the output voltage [Fig. 4.42(b)]. Note that if $I_1 = 0$, then $V_{out} = +g_m R_D$, and if $I_2 = 0$, then $V_{out} = -g_m R_D$. For $I_1 = I_2$, the gain drops to zero.

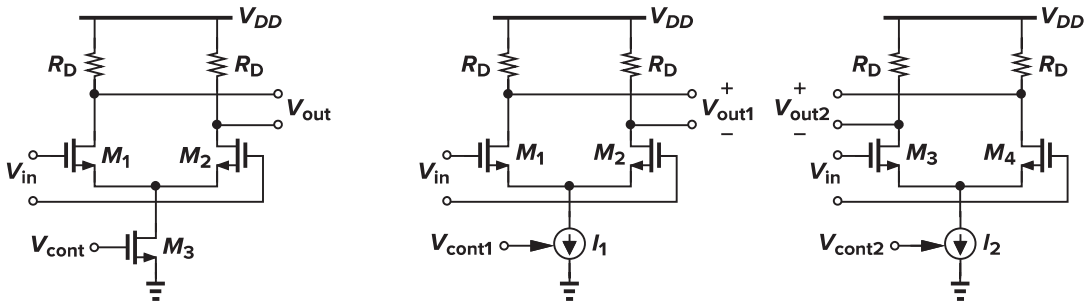


Figure 4.41 (a) Simple VGA; (b) two stages providing variable gain.

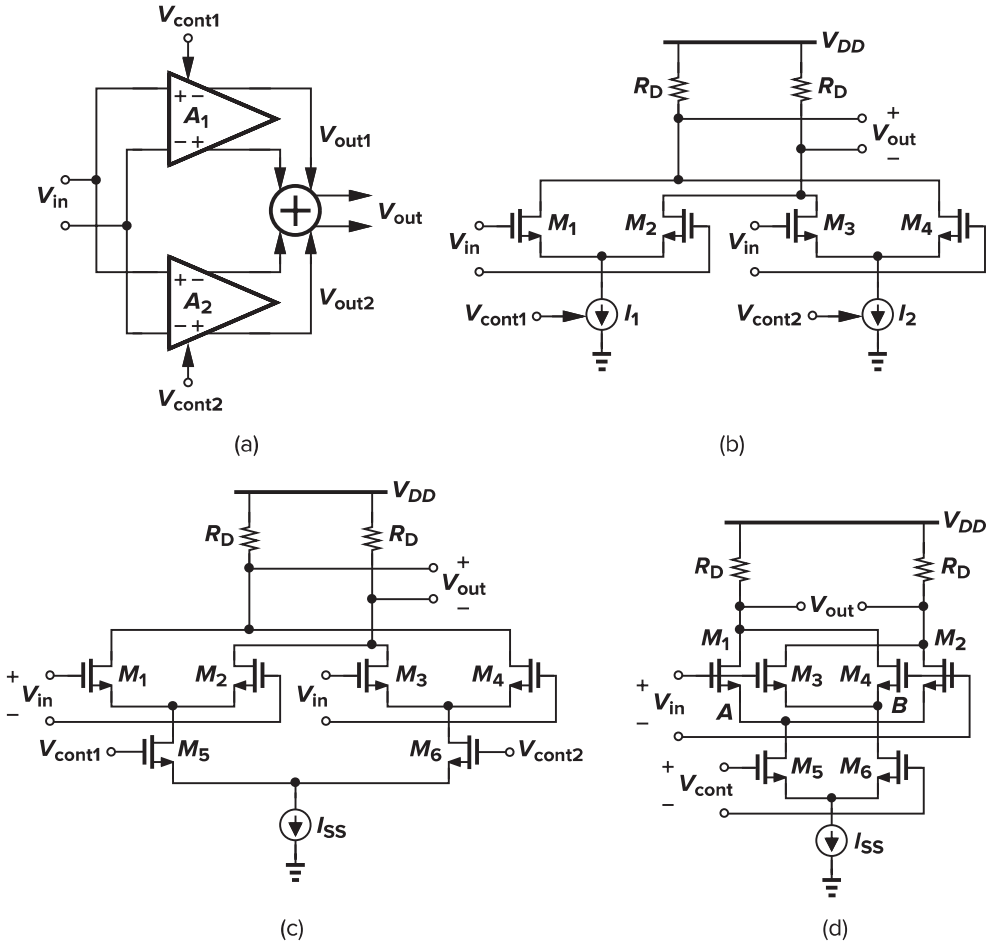


Figure 4.42 (a) Summation of the output voltages of two amplifiers; (b) summation in the current domain; (c) use of M_5-M_6 to control the gain; (d) Gilbert cell.

In the circuit of Fig. 4.42(b), V_{cont1} and V_{cont2} must change I_1 and I_2 in opposite directions such that the gain of the amplifier changes monotonically. What circuit can vary two currents in opposite directions? A differential pair provides such a characteristic, leading to the topology of Fig. 4.42(c). Note that for a large $|V_{cont1} - V_{cont2}|$, all of the tail current is steered to one of the top differential pairs and the gain from V_{in} to V_{out} is at its most positive or most negative value. If $V_{cont1} = V_{cont2}$, the gain is zero. For simplicity, we redraw the circuit as shown in Fig. 4.42(d). Called the "Gilbert cell" [2], this topology is widely used in many analog and communication systems. In a typical design, M_1–M_4 are identical, and so are M_5 and M_6.

▶ Example 4.13

Explain why the Gilbert cell can operate as an analog voltage multiplier.

Solution

Since the gain of the circuit is a function of $V_{cont} = V_{cont1} - V_{cont2}$, we have $V_{out} = V_{in} \cdot f(V_{cont})$. Expanding $f(V_{cont})$ in a Taylor series and retaining only the first-order term, αV_{cont}, we have $V_{out} = \alpha V_{in} V_{cont}$. Thus, the circuit can multiply voltages. This property accompanies any voltage-controlled variable-gain amplifier. ◀

As with a cascode structure, the Gilbert cell consumes a greater voltage headroom than a simple differential pair does. This is because the two differential pairs M_1–M_2 and M_3–M_4 are "stacked" on top of the control differential pair. To understand this point, suppose the differential input, V_{in}, in Fig. 4.42(d) has a common-mode level $V_{CM,in}$. Then, $V_A = V_B = V_{CM,in} - V_{GS1}$, where M_1–M_4 are assumed identical. For M_5 and M_6 to operate in saturation, the CM level of V_{cont}, $V_{CM,cont}$, must be such that $V_{CM,cont} \leq V_{CM,in} - V_{GS1} + V_{TH5,6}$. Since $V_{GS1} - V_{TH5,6}$ is roughly equal to one overdrive voltage, we conclude that the control CM level must be lower than the input CM level by at least this value.

In arriving at the Gilbert cell topology, we opted to vary the gain of each differential pair through its tail current, thereby applying the control voltage to the bottom pair and the input signal to the top pairs. Interestingly, the order can be exchanged while still obtaining a VGA. Illustrated in Fig. 4.43(a), the idea is to convert the input voltage to current by means of M_5 and M_6 and route the current through M_1–M_4 to the output nodes. If, as shown in Fig. 4.43(b), V_{cont} is very positive, then only M_1 and M_3 are on and $V_{out} = g_{m5,6} R_D V_{in}$. Similarly, if V_{cont} is very negative [Fig. 4.43(c)], then only M_2 and M_4 are on and $V_{out} = -g_{m5,6} R_D V_{in}$. For a zero differential control voltage, $V_{out} = 0$. The input differential pair may incorporate degeneration to provide a linear voltage-to-current conversion.

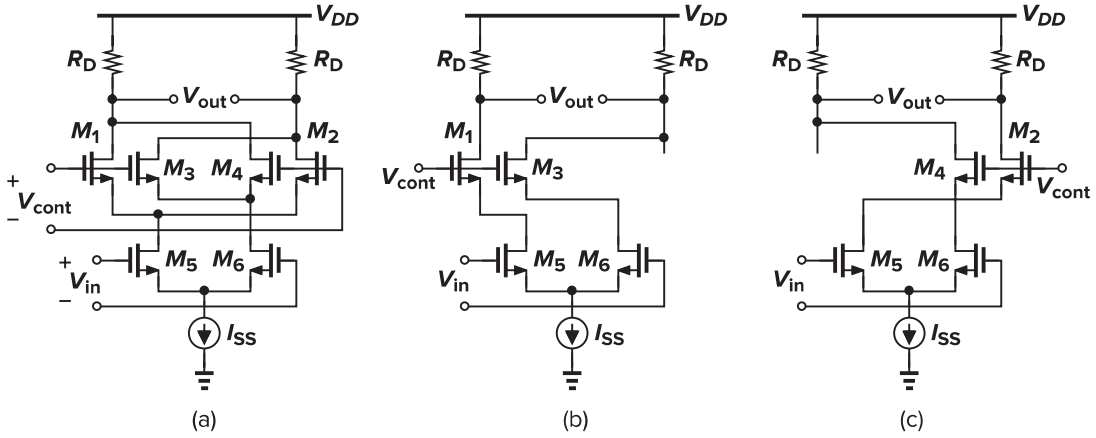


Figure 4.43 (a) Gilbert cell sensing the input voltage by the bottom differential pair; (b) signal path for very positive V_{cont}; (c) signal path for very negative V_{cont}.

References

[1] P. R. Gray and R. G. Meyer, *Analysis and Design of Analog Integrated Circuits*, 3d ed. (New York: Wiley, 1993).
[2] B. Gilbert, "A Precise Four-Quadrant Multiplier with Subnanosecond Response," *IEEE J. Solid-State Circuits*, vol. SC-3, pp. 365–373, Dec. 1968.

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3$ V where necessary. All device dimensions are effective values and in microns.

4.1. Suppose the total capacitance between adjacent lines in Fig. 4.2 is 10 fF and the capacitance from the drains of M_1 and M_2 to ground is 100 fF.
(a) What is the amplitude of the glitches in the analog output in Fig. 4.2(a) for a clock swing of 3 V?
(b) If in Fig. 4.2(b), the capacitance between L_1 and L_2 is 10% less than that between L_1 and L_3, what is the amplitude of the glitches in the differential analog output for a clock swing of 3 V?

4.2. Sketch the small-signal differential voltage gain of the circuit shown in Fig. 4.9(a) if V_{DD} varies from 0 to 3 V. Assume that $(W/L)_{1-3} = 50/0.5$, $V_{in,CM} = 1.3$ V, and $V_b = 1$ V.

4.3. Construct the plots of Fig. 4.9(c) for a differential pair using PMOS transistors.

4.4. In the circuit of Fig. 4.11, $(W/L)_{1,2} = 50/0.5$ and $I_{SS} = 0.5$ mA.
(a) What is the maximum allowable output voltage swing if $V_{in,CM} = 1.2$ V?
(b) What is the voltage gain under this condition?

4.5. A differential pair uses input NMOS devices with $W/L = 50/0.5$ and a tail current of 1 mA.
(a) What is the equilibrium overdrive voltage of each transistor?
(b) How is the tail current shared between the two sides if $V_{in1} - V_{in2} = 50$ mA?
(c) What is the equivalent G_m under this condition?
(d) For what value of $V_{in1} - V_{in2}$ does the G_m drop by 10%? By 90%?

4.6. Repeat Problem 4.5 with $W/L = 25/0.5$ and compare the results.

4.7. Repeat Problem 4.5 with a tail current of 2 mA and compare the results.

4.8. Sketch I_{D1} and I_{D2} in Fig. 4.19 versus $V_{in1} - V_{in2}$. For what value of $V_{in1} - V_{in2}$ are the two currents equal?

4.9. Consider the circuit of Fig. 4.32, assuming $(W/L)_{1,2} = 50/0.5$ and $R_D = 2$ kΩ. Suppose R_{SS} represents the output impedance of an NMOS current source with $(W/L)_{SS} = 50/0.5$ and a drain current of 1 mA. The input signal consists of $V_{in,DM} = 10\ \text{mV}_{pp}$ and $V_{in,CM} = 1.5\ \text{V} + V_n(t)$, where $V_n(t)$ denotes noise with a peak-to-peak amplitude of 100 mV. Assume that $\Delta R/R = 0.5\%$.
(a) Calculate the output differential signal-to-noise ratio, defined as the signal amplitude divided by the noise amplitude.
(b) Calculate the CMRR.

4.10. Repeat Problem 4.9 if $\Delta R = 0$, but M_1 and M_2 suffer from a threshold voltage mismatch of 1 mV.

4.11. Suppose the differential pair of Fig. 4.37(a) is designed with $(W/L)_{1,2} = 50/0.5$, $(W/L)_{3,4} = 10/0.5$, and $I_{SS} = 0.5$ mA. Also, I_{SS} is implemented with an NMOS device having $(W/L)_{SS} = 50/0.5$.
(a) What are the minimum and maximum allowable input CM levels if the differential swings at the input and output are small?
(b) For $V_{in,CM} = 1.2$ V, sketch the small-signal differential voltage gain as V_{DD} goes from 0 to 3 V.

4.12. In Problem 4.11, suppose M_1 and M_2 have a threshold voltage mismatch of 1 mV. What is the CMRR?

4.13. In Problem 4.11, suppose $W_3 = 10$ μm, but $W_4 = 11$ μm. Calculate the CMRR.

4.14. For the differential pairs of Fig. 4.37(a) and (b), calculate the differential voltage gain if $I_{SS} = 1$ mA, $(W/L)_{1,2} = 50/0.5$, and $(W/L)_{3,4} = 50/1$. What is the minimum allowable input CM level if I_{SS} requires at least 0.4 V across it? Using this value for $V_{in,CM}$, calculate the maximum output voltage swing in each case.

4.15. In the circuit of Fig. 4.39(a), assume that $I_{SS} = 1$ mA and $W/L = 50/0.5$ for all the transistors.
(a) Determine the voltage gain.
(b) Calculate V_b such that $I_{D5} = I_{D6} = 0.8(I_{SS}/2)$.
(c) If I_{SS} requires a minimum voltage of 0.4 V, what is the maximum differential output swing?

4.16. Assuming that all the circuits shown in Fig. 4.44 are symmetric, sketch V_{out} as **(a)** V_{in1} and V_{in2} vary differentially from zero to V_{DD}, and **(b)** V_{in1} and V_{in2} are equal and vary from zero to V_{DD}.

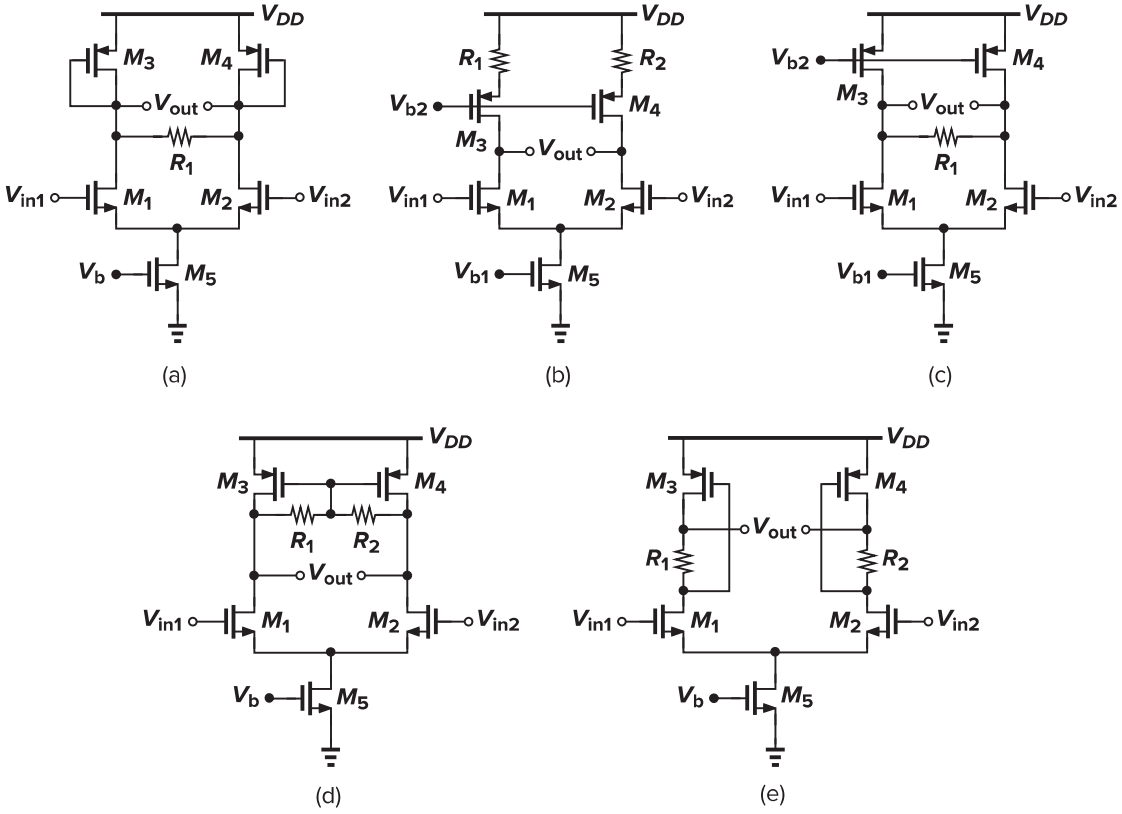


Figure 4.44

4.17. Assuming that all the circuits shown in Fig. 4.45 are symmetric, sketch V_{out} as **(a)** V_{in1} and V_{in2} vary differentially from zero to V_{DD}, and **(b)** V_{in1} and V_{in2} are equal and vary from zero to V_{DD}.

4.18. Assuming that all the transistors in the circuits of Figs. 4.44 and 4.45 are saturated and $\lambda \neq 0$, calculate the small-signal differential voltage gain of each circuit.

4.19. Consider the circuit shown in Fig. 4.46.
(a) Sketch V_{out} as V_{in1} and V_{in2} vary differentially from zero to V_{DD}.
(b) If $\lambda = 0$, obtain an expression for the voltage gain. What is the voltage gain if $W_{3,4} = 0.8W_{5,6}$?

4.20. For the circuit shown in Fig. 4.47,
(a) Sketch V_{out}, V_X, and V_Y as V_{in1} and V_{in2} vary differentially from zero to V_{DD}.
(b) Calculate the small-signal differential voltage gain.

4.21. Assuming no symmetry in the circuit of Fig. 4.48 and using no equivalent circuits, calculate the small-signal voltage gain $(V_{out})/(V_{in1} - V_{in2})$ if $\lambda = 0$ and $\gamma \neq 0$.

4.22. Due to a manufacturing defect, a large parasitic resistance has appeared between the drain and source terminals of M_1 in Fig. 4.49. Assuming $\lambda = \gamma = 0$, calculate the small-signal gain, common-mode gain, and CMRR.

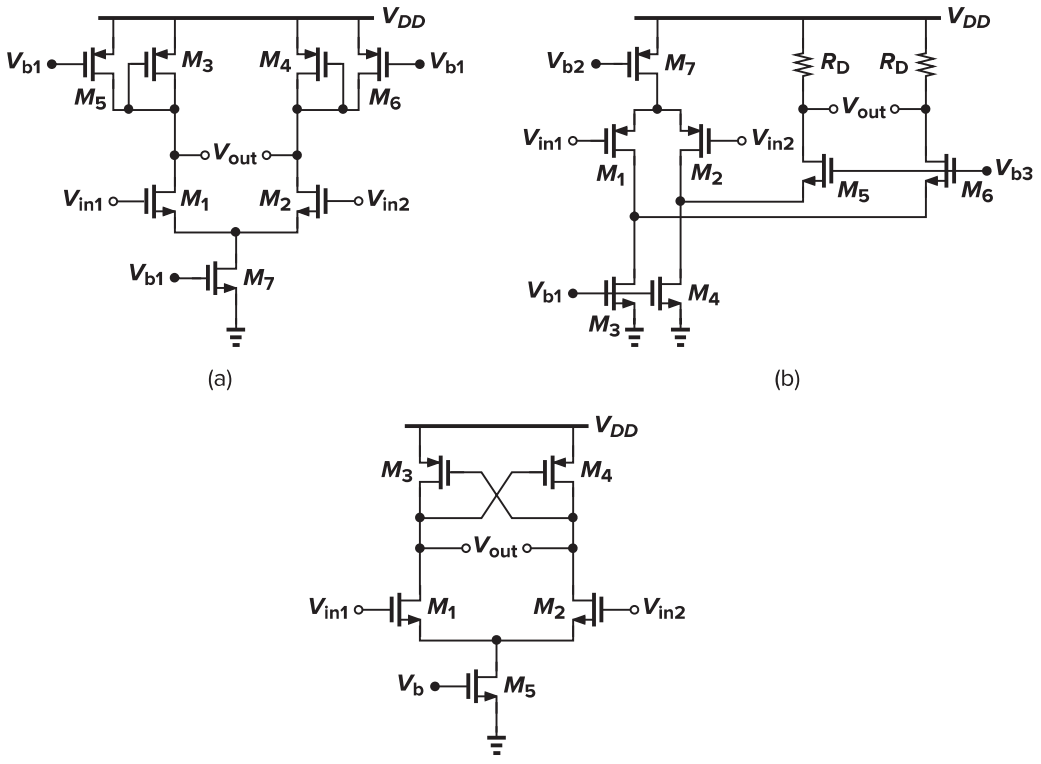


Figure 4.45

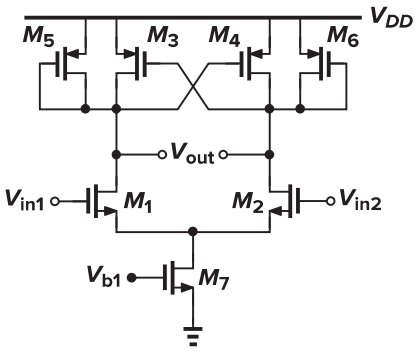


Figure 4.46

4.23. Due to a manufacturing defect, a large parasitic resistance has appeared between the drains of M_1 and M_4 in the circuit of Fig. 4.50. Assuming $\lambda = \gamma = 0$, calculate the small-signal gain, common-mode gain, and CMRR.

4.24. In the circuit of Fig. 4.51, all of the transistors have a W/L of 50/0.5, and M_3 and M_4 are to operate in the deep triode region with an on-resistance of 2 kΩ. Assuming that $I_{D5} = 20\ \mu$A and $\lambda = \gamma = 0$, calculate the input common-mode level that yields such resistance. Sketch V_{out1} and V_{out2} as V_{in1} and V_{in2} vary differentially from 0 to V_{DD}.

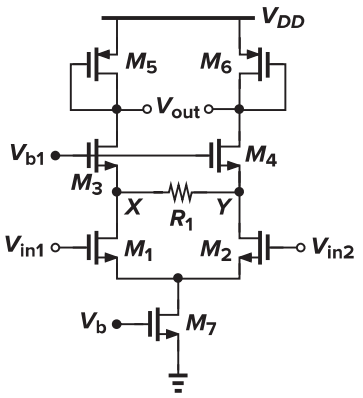


Figure 4.47

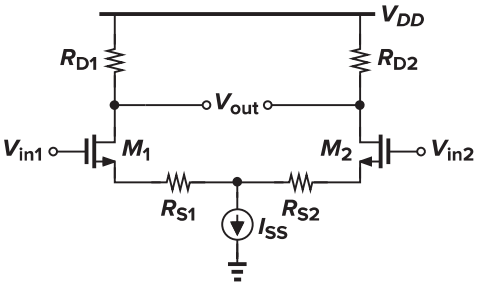


Figure 4.48

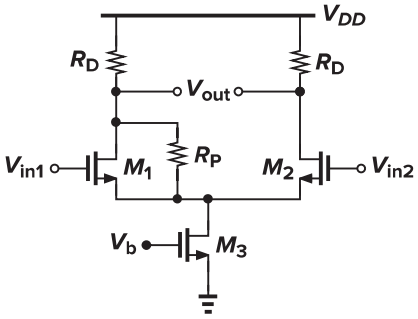


Figure 4.49

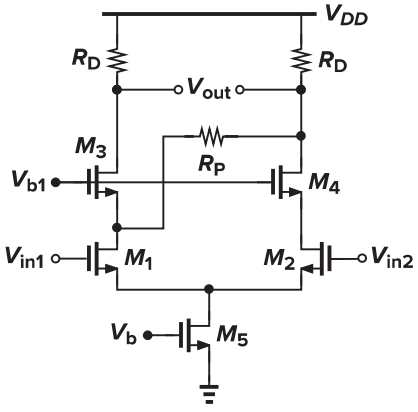


Figure 4.50

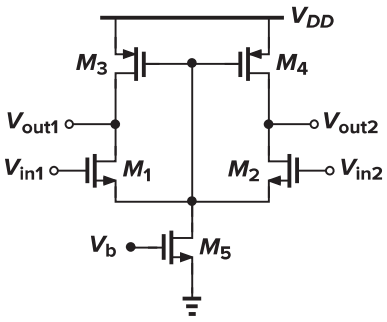


Figure 4.51

4.25. In the circuit of Fig. 4.37(b), $(W/L)_{1-4} = 50/0.5$ and $I_{SS} = 1$ mA.
(a) What is the small-signal differential gain?
(b) For $V_{in,CM} = 1.5$ V, what is the maximum allowable output voltage swing?

4.26. In the circuit of Fig. 4.39, assume that M_5 and M_6 have a small threshold voltage mismatch of ΔV and I_{SS} has an output impedance R_{SS}. Calculate the CMRR.

4.27. What happens if R_{SS} in Eq. (4.56) becomes very large? Can we obtain the same result by analyzing a differential pair having an ideal tail current source but $g_{m1} \neq g_{m2}$?

4.28. In Example 4.5, how much input dc imbalance can be tolerated if the small-signal gain must not drop by more than 5%?

4.29. In the lemma illustrated in Fig. 4.20, suppose channel-length modulation is not neglected. Assuming the two devices are connected to two equal load resistors, explain intuitively why the lemma still holds.

4.30. Does the lemma in Fig. 4.20 still hold if the devices have body effect? Explain.

4.31. Repeat Example 4.7 using Method I.

4.32. Prove the lemma illustrated in Fig. 4.20 if the tail current source is replaced by a resistor R_T.

4.33. What happens to the plots on Fig. 4.13 as W/L increases? Determine the area under the G_m plot and use the result to explain why the peak G_m must increase as W/L increases.

4.34. Assuming that I_1 and I_{SS} in Fig. 4.52 are ideal and $\lambda, \gamma > 0$, determine V_{out1}/V_{in} and V_{out2}/V_{in}.

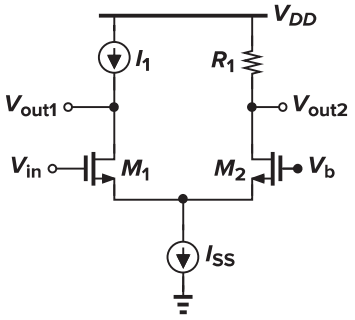


Figure 4.52

4.35. In Problem 4.11, suppose M_3 and M_4 have a threshold voltage mismatch of 1 mV. Calculate the CMRR.